Study Questions, by Topic (cumulative), will be added here as the term progresses

I. Introduction

1. For each of the following statements, explain whether they are normative or positive

a) Government expenditure on road construction should decrease unemployment rates.

Normative/Positive	**Explanation**
Positive	It is a statement of presumed fact. There is no value judgement. The word should is used in a descriptive rather than proscriptive manner.

b) Government expenditure on road construction is predicted to increase unemployment rates.

Normative/Positive	**Explanation**
Positive	It is a statement of presumed fact. A prediction from some model. No value judgement.

c) The government of Vulcan spends too much on road construction.

Normative/Positive	**Explanation**
Normative	It is a statement that implies judgement. It is proscribing what is good/not good.

d) The temperature today is too warm for it to snow.

Normative/Positive	**Explanation**
Positive	Too warm is not intended as a judgement in this context. It is a predictive statement.

e) There is too little snow fall (.2 cm) to justify salting the roads.

Normative/Positive	Explanation
Normative	This statement implies a judgement as to whether salting the roads is justified. It is proscriptive making a judgement.

f) Canadian GDP is currently growing at 8% per year.

Normative/Positive	Explanation
Positive	This is a descriptive statement that sounds factual. It is incorrect, but it is a positive statement because it describes rather than proscribes/judges.

g) Monetary policy needs to focus on inflation to be effective.

Normative/Positive	Explanation
Normative	This statement proscribes what policy is more effective. It is a value judgement rather than a descriptive statement (as the most "effective" policy is not based on any universal/objective description here).

2. For each of the normative statements in question 1, transform those statements into positive ones (you may/not need as many rows as I have provided) **Many transformations are acceptable.**

Original Statement (normative):	Transformed Statement (positive):
Monetary policy needs to focus on inflation to be effective.	Monetary policy is currently focused on inflation.
There is too little snow fall (.2 cm) to justify salting the roads.	The snow fall (.2 cm) does not meet the region's set criteria for salting the roads.
The government of Vulcan spends too much on road construction.	The government of Vulcan spends 30% of GDP on road construction.

3. For each of the positive statements in question 1, transform those statements into normative ones (you may/not need as many rows as I have provided) **Many transformations are acceptable.**

Original Statement (positive):	**Transformed Statement (normative):**
Government expenditure on road construction should decrease unemployment rates.	Government expenditure on road construction is a great policy to decrease unemployment rates.
Government expenditure on road construction is predicted to increase unemployment rates.	Government expenditure on road construction is an effective policy to increase unemployment rates
The temperature today is too warm for it to snow.	The temperature today is too warm.
Canadian GDP is currently growing at 8% per year.	Canadian GDP is not growing fast enough, at only 8% per year

4. Suppose you publish a research report indicating that more generous (larger) Employment Insurance payments will reduce workers' incentive to find new jobs quickly, which will result in higher unemployment rates.

a) Is this reported outcome written in a normative or positive way? Explain.

Normative/<u>Positive</u> (underline one)	**Explanation:** You are stating presumed facts in a descriptive manner, with no value judgement. Therefore it is written in a positive way

b) Is there any reason why you might want to use some normative statements when describing the implications of your research?

<u>Yes</u>/No (underline one)	**Explanation:** Readers will assume your intent, will assume your value judgements when reading your research. It is better to be clear and specific about your normative view, so that others do not infer a message that you did not intend with your research.

c) What additional statement(s) might you wish to make in this context? **Many good answers possible**

Additional Statement(s):
While more generous EI is predicted to increase unemployment rates due to prolonged search. Job matches coming out of a longer search are observed to be better matches and more productive. Therefore, generous EI can be beneficial for long run economic growth. While more generous EI is predicted to increase unemployment rates, it provides better health outcomes for workers and their families. A generous EI could be coupled with better job search supports to counteract the negative employment effects of higher EI payments.

5. Suppose that you are working as a policy advisor for the Government of Rohan, and you design a model to predict the implications of minimum wage increases on employment levels. Suppose that the model you design is a market clearing model that assumes all workers are identical and all firms are identical. Your model predicts that when minimum wage is set above the market clearing wage, employment levels (N) falls. Your model also predicts that when minimum wage is above the market clearing wage that unemployment will rise, but when minimum wage is at or below the market clearing wage, that unemployment will be zero.

a) Using the Model Evaluation criteria from our lecture slides, explain whether your model meets each criterion: "Well", "Somewhat", or "Not at All"

Criterion	**Meets criterion "Well," "Somewhat," "Not at All"**	**Explanation**
Realistic Assumptions	Somewhat	Market clearing is not the best assumption to make for this analysis because it can occasionally predict unemployment rates of 0, which we know does not realistically capture the labour market. Since the research focus was aimed on Employment rather than Unemployment, and the model is able to predict decently there, the assumptions realistic enough for those purposes. Likewise, assuming homogeneous workers and firms is not very realistic, but for the purposes of minimum wage analysis, may be sufficient – at least for a qualitative analysis – to provide a generally useful depiction of the broad labour market functions.
Understandable & Manageable	Well	This is a very simple model, and therefore easier to understand, and very easy to solve.
Testable Implications with Available Data	~Well	Data on minimum wage is publicly available. Data on employment is also publicly available. Both are regularly tracked, and therefore it is easy to obtain long series of standardized data with which to test the hypothesis on the effects of minimum wage hikes on employment levels. It is not perfect, in the sense that there may not be sufficient data to estimate a causal relationship (but you do not have that info yet).
Predictions Consistent with Data	Somewhat	The prediction of employment falling with minimum wage is generally consistent (although this is simple correlation without causation being identified). The prediction of zero unemployment is, of course, not consistent with the data.

b) Based on the evaluation of your model against these criterion, is your model a reasonable one to advise policy makers in the Government of Rohan on the implications of minimum wage on employment levels? Explain why or why not.

Reasonable? **<u>Yes</u>/No/Uncertain** **(underline one)**	**Explanation**: This is a subjective question, but generally speaking, if your model is somewhat meeting the criterion, and in the absence of strong concerns, the model is reasonable to provide some advice policy makers.

c) Would you recommend considering other models for this policy question? Why or why not?

Other models? **<u>Yes</u>/No** **(underline one)**	**Explanation:** Yes. I would recommend considering non-market clearing models which can better capture the implications on employment while generating more consistent predictions for unemployment as well.

6. Explain one benefit of using algebraic expressions or mathematical characterizations of behaviour/economic structure/constraints as we tend to do in theoretical analysis in Economics.

Benefit:	*discipline our thinking, & maintaining consistency in our characterization of behaviours. *help us keep track of feedback (multiplier) effects.
Explanation:	*mathematical characterizations help us to discipline our thinking as we aim to understand the implications of behaviour, structure and constraints on economic outcomes. With algebraic expressions, we will be consistently characterizing behaviours. *these characterizations (structured equations) help us remember, but also help us keep track of feedback (multiplier) effects when outcomes in one market affect those of another and then feedback into the original market.

II. Measurement, Accounting and Data

7. Suppose you are hired by the government to assess the impact of the latest (COVID) recession on household well-being using detailed GDP measures from Statistics Canada over a 5 year period.

a) Which method(s) of GDP calculation would you use, and why?

Method(s):	I would use both Income and Expenditure Approaches
Explanation:	The income approach will enable the research to investigate the impact of the recession on household earnings, which are important as a measure of financial sufficiency and stability. The expenditure approach will enable the research to investigate the impact of the recession on consumption, which is an important measure of material well-being. Consumption is a good measure to capture immediate wellbeing, but income captures long-term financial well-being. It is worthwhile to note that high consumption cannot be sustained long-term and will reduce saving (and deplete wealth) in the absence of sufficient income. The two are related, but not identical measures of household well-being as consumption does not always move 1:1 with income.

b) Would you use Real or Nominal measures of GDP, and why?

Real or Nominal:	Real GDP
Explanation:	Because of the high inflation over this recession, using Real GDP will be particularly important, even though it is a short time frame. It may be illuminating to contrast with nominal GDP because of the disproportionate growth of wages relative to inflation near the latter years.

c) Which inflation index would you use and what are the drawbacks of this measure for your task?

Index:	CPI, chain weighted
Explanation:	Because we are looking at household well-being, using the index which focuses on key household expenditure would b appropriate. However, we would want to use the chain weighted variant, because with such large changes in inflation and consumption patterns over 5 years, the choice of base year may dramatically alter estimates. A drawback of the CPI is that the basket weights may not do a great job of capturing the value of items which changed dramatically over this period, and a drawback of the chain weighting is that it is more complicated to assess the total change vs change in sub-sets or categories of GDP

Note: for subjective questions, more than one answer may be accepted, and the explanation will be very important in determining your marks.

8. Explain whether the items below will be included in Canada's GDP, GNP, both, or uncertain:

a. Tuition expenditures of Irish students in Canada

Answer: Both
Explanation: Tuition expenditures of Irish students in Canada are included in GDP because they are payment for services produced in Canada. These expenditures are also part of GNP because they are payments to domestic factors of production (staff and faculty of Canadian Universities)

b. Canadian singer, Anne Murray's, concert in Vienna

Answer: GNP only
Explanation: Anne Murray is earning income from a good (her concert) that she produces abroad. Because it is not produced domestically, it is not included in GDP; but because it is produced by a domestic factor of production it is included in GNP

c. Louise Penny's sale of her latest Three Pines Quebec Mystery Novel

Answer: Uncertain
Explanation: The author is Canadian living and writing in Canada, but the book may be printed internationally – so it depends on the sale. But if we are referring just to Louise Penny's sale of the book to her publisher (which is her value added) would be part of <u>both</u> GDP and GNP.

9. One of your class mates claims that Gross Domestic Product is an unreliable measure of aggregate economic activity. She says that GDP will underestimate actual economic activity because much intangible production is not included. She further claims that exclusion of the underground economy (drug sales, prostitution income) also aggravates the underestimation.

a. Is it possible that GDP could underestimate economic activity? Explain

Answer: Yes/No	Yes
Explanation	She is correct, not only for the reasons she listed (e.g. Brand and market niche intangible asset production are not included in GDP), but also home production is not included in GDP, and represents a substantial amount of production that is likely changing over time.

b. Explain why GDP may actually be over-estimated, rather than underestimated.

Explanation	GDP may be over-estimated because it does not take into account the depletion of natural resources, nor does it take into account pollution. GDP may also not accurately reflect improved wellbeing/health/happiness (whether it is under or overestimating these).

10. For each of the following scenarios, will our measure of GDP over-estimate or under-estimate aggregate economic activity. Explain.

a) Cleaning your apartment/dorm room. (2)

Answer: Under-estimate

Explanation: You are 'producing' a clean room. If you paid a housekeeper, it would have been included in GDP, but because you did it yourself, it is not included in GDP. Thus GDP under-estimates true aggregate economic activity.

b) Cleaning up municipal water that had been contaminated from agricultural run-off. (2)

Answer: Over-estimate

Explanation: Clean-up of the water does not account for the loss of the clean resource that we had in the first place. We are not producing new clean water, we are repairing the cleanliness of the water we had to begin with.

c) Taking your cousin to pet the Llamas in Waterloo Park. (2)

Answer: Under-estimate

Explanation: You are 'producing' happiness in this leisure activity, but the value of this production is not included in GDP. Thus GDP under-estimates true aggregate economic activity.

11. The Mrs Ws sell teleportation devices which they purchase from a manufacturer for $100,000. The Mrs Ws pay their employees a total of $40,000, pay $25,000 in taxes and have profits of $5,000. What is the value added to the economy by the Mrs Ws? Show your work.

Value Added: 70,000

Calculation: Using the information above, we know that the W's have sold a total of
$100,000+ $40,000 + $25,000 +$5,000=$170,000 (if their profits are $5,000
and their costs are $165,000, then their total sales must be $170,000)
The intermediate goods are valued at $100,000. $170,000-100,000=$70,000.
Note that this amount is also the amount of income generated by the W's.
Value added= income generated = wages + taxes + profits = $70,000

12. Answer the following questions and provide explanations for each answer:

a. Statistics Canada adopted chain weighting in May 2001. This change led to a downward revision in investment rates (the rate of change of the capital stock), because the fall in computer prices gives the growth in the volume of computing equipment a smaller weight when more recent prices are used to construct real investment I. What effect will this revision have on measured TFP growth in Canada?

Effect on measured TFP Growth:	An upward revision of TFP growth
Explanation: Recall that investment is growth of K, since A is usually calculated from values of Y, N, and K using the production function: A = Y/F(K,N) where F(K,N) is increasing in K, then a downward revision of investment rates (a downward revision of the change in K) would mean that the denominator growth would be revised downward. A smaller denominator leads to higher ratio. So the revision of the growth of A would be higher.	

b. Anyanwu's medicinal flower shop sells $40 worth of flowers (including taxes), to it's customers, and pays $10 to the farmer for the flowers, $5 to the government in taxes, and $20 to a retail clerk. The farmer, in turn, pays a farm hand $2 to cut the flowers and pays $1 in taxes. The retail clerk pays $6 in income tax, while the farm hand pays $0 in income tax. Use the income approach to calculate GDP from Anyanwu's. Show your calculations.

GDP using Income Approach:	40
Calculations: taxes=$6 + $5 + $1 + $0=$12 wages=$20-6 + $2 = $16 profits=$40-10-5 -20 + $10-2-1=$12 GDP=12+16+12=40	

c. If Government debt started at zero and for 10 years government expenditure exceeds revenue by $4, calculate the government deficit in year 10 and the government debt at the end of year 5.

Deficit in year 10:	$4 (deficit is a flow variable and is the difference between expenditure and revenue over a defined period of time – in this case, over a year)
Debt in year 5:	$20 (debt is a stock variable, and is measured as the cumulated amount at a single point in time, in this case, at the end of year year 5. ($4/year for 5 years is 20)

13. Given the following information about Spooky Island that sells jars of Protoplasm

	Year 1	Year 3	Year 3
Quantity of Protoplasm Jars Sold	400	435	500
Price of Protoplasm Jars	1.00	1.15	1.20

a. Calculate Nominal GDP and the GDP deflator for years 2 and 3 assuming year 1 is the base year. Explain and show your work.

	Year 2	**Year 3**
Nominal GDP	500.25	600
GDP Deflator	1.15	1.20

Calculations:

Nominal GDP is the value of output in current dollars, so for year 2, nominal GDP is 435x1.15 = 500.25. In year 3 nominal GDP is 500x1.20=600

The GDP deflator is Nominal/Real GDP, and since year 1 is our base year, and P=1 in year 1, then real GDP for years 2 and 3 are simply the quantity sold. This means the deflator for year 2 is 500.25/435 = 1.15, and the deflator for year 3 is 600/500 = 1.20

b. Calculate the inflation from year 2 to 3. Show your work.

Inflation year 2 to 3:	4%

Calculation:

Inflation= $(P_{t+1} – P_t)/ P_t$

Since there is only one good sold, this calculation is:

Inflation=(1.2-1.15)/1.15 =0.04 or 4%

Note that this is also the percentage change in the GDP deflator over years 2 to 3.

c. Assume jars of protoplasm do not rot from one year to the next. Also assume that you don't care whether you consume your jars of protoplasm in year 2 or 3. If the nominal interest rate in year 2 is 4%, would you be better off loaning $115 in year 2, borrowing $115 in year 2, or be equally well off regardless? Why?

Better off loaning/borrowing or equally well off:	Equally well off whether borrowing or lending
Explanation: Because i=0.04 and π=0.04, then r=0.04-0.04=0 With a real interest rate of zero, you gain no additional real goods in year 3 by lending, nor loose any real goods by borrowing. So if you had $115 in year 2, you could buy 100 jars of protoplasm , or you could lend the money out and receive $120 in year 3, with which you could also buy 100 jars of protoplasm. Since you don't care which year you consume the jars of protoplasm, you are equally well off lending or not. The same is true for borrowing.	

d. Now suppose that nominal interest rates fall to 3%, but jars of protoplasm go bad if you don't consume them right away, and you don't have capacity to consume more than 100 jars a year. If you had $230 in year 2, and spent 115 of it on jars in year 2. Would you be better off loaning your remaining $115 in year 2, or saving the cash in your pocket? Explain.

Better off lending or keeping in pocket:	Better off lending
Explanation: Because i=0.03 and π=0.04, then r=0.03-0.04=-1 With a real interest rate <0, your real spending power in year 3 is lower. However even though the interest rate is negative, then lending out your remaining 115 is better than just keeping 115 in your pocket. This is because in year 3, you will have 115*1.03 = 118.45, with which you can buy 98.7 jars, which is better than the 95.8 jars that you could buy with only 115. (note: you weren't given the option to borrow here, partly because there is no point—you can't consume more than 100 in year 1 and the jars of protoplasm go bad. Suppose you borrowed $10 anyway, next year you would have to pay back 13, so it would deplete 3 from the 115 you would have otherwise have –in other words, no point. If you borrowed 10 to invest, but faced the same interest rate that you paid, 3%, you would be no better off. So again, no point)	

III. Economic Frameworks

14. Suppose that you wish to design a model that would allow policy makers to predict the impact on unemployment that would arise from an increase in the compensation for lost earnings (EI payments) under the Employment Insurance program.

a) Explain the type of model you would like to use, according to the following classifications: static versus dynamic, general versus partial equilibrium, and a market clearing versus a non-market clearing model. Be sure to defend your choices.

Choice (underline one)	**Explanation:**
Static vs Dynamic	Dynamic because unemployment, and EI has a time dimension to it. More generous EI payments mean you may be able to subsist longer and take more time to search for a job (not a bad thing). Therefore, unemployment rates would be higher due to the greater time in unemployment.
General vs Partial	Whenever you wish to analyze a policy outcome, it is best to use a General Equilibrium model (although we'll prefer non-equilibrium) because you want to assess the total impact, inclusive of all feedback loops (e.g. if unemployment persists because of generous EI, consumption demand might fall, recessionary and reducing demand for labour)
Market Clearing vs non-clearing	To study unemployment, you will want to use a non-market clearing model, because with market clearing $N^S=N^D$ and unemployment is always zero

b) Would unemployment be an exogenous or endogenous variable in your model? Explain.

Unemployment Endogenous or Exogenous:	Endogenous
Explanation:	Because unemployment is determined within (predicted by) our model – it is an outcome of our model, it is an endogenous variable.

15. Suppose an economy has the production function $Y = AK^{0.25}N^{0.75}$
In 2021: Real GDP = 74,802; K = 268,147; N = 2414
In 2022: Real GDP = 78,561; K = 301,262; N = 2632

a. Calculate total factor productivity (TFP) for 2021 and 2022.

TFP 2021: 9.54	**TFP 2022:** 9.54
Calculations: TFP = A and $A = Y/ (K^{0.25}N^{0.75})$ $A(2021) = 74{,}802/ (268{,}147^{0.25} x 2414^{0.75}) = 9.54$ $A(2022) = 78{,}561/ (301{,}262^{0.25} x 2632^{0.75}) = 9.12$	

b. What is the growth of total factor productivity?

Growth of TFP:	-4%
Calculation: TFP growth, growth of A = 100 * [A(2002) - A(2001)] / A(2001) = 100 * [9.12 - 9.54] / 9.54 = -4% or -0.04	

c. Calculate the percent increase in real GDP from 2021 to 2022.

Percentage increase in real GDP:	5%
Calculation: GDP growth = 100 * [GDP(2022) - GDP(2021)] / GDP(2021) = 100 * [78,561 - 74,802] / 74,802 = 5% or 0.05	

d. Suppose productivity remained constant from 2022 to 2023, but K and N both triple during that time. What will GDP be in 2023?

GDP:	235,683
Calculation: $Y= A (K^{0.25}N^{0.75})$ In 2023 $Y=A (3*K)^{0.25}(3*N)^{0.75}$ --> $Y=A*3^{0.25}*K^{0.25} *3^{0.75} *N^{0.75} = 3(A(K^{0.25}N^{0.75})) = 3Y = 3* 78{,}561 = 235{,}683$ Because this function exhibits constant returns to scale (the powers sum to 1), then any constant multiplied by both input factors, is equivalent to multiplying output by that constant. Note: if you just triple one input, output will NOT triple: e.g. $A(K^{0.25}(3N)^{0.75})=3^{0.75}(A(K^{0.25}N^{0.75}))=3^{0.75}Y \neq 3Y$	

16. Suppose the country of Narnia has the following output and input levels for 2023 and 2024:

Year	Y	N	K
2023	20	1	4
2024	1000	4	25

a) If Economists believe that the production function for Narnia's economy is $Y=AKN^{0.5}$, what is TFP growth for 2023-2024?

TFP Growth: 300% (or 3)
Calculation: $A=Y/KN^{0.5}$ $A_{2010} = 20/4 = 5$, $A_{2011} = 1000/50 = 20$ Growth=100* (N-0)/0 = 100*(20-5)/5 = 300%

b) Now suppose that the true production process in Narnia is characterized by $Y=AK^{0.5}N^{0.5}$. Calculate actual TFP growth.

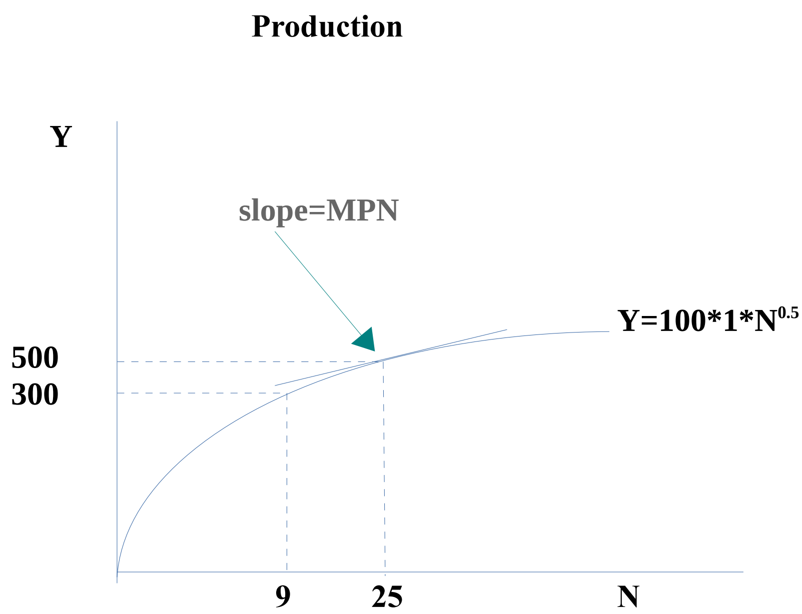
TFP Growth: 900% (or 9)
Calculation: $A=Y/K^{0.5}N^{0.5}$ $A_{2010} = 20/2 = 10$, $A_{2011} = 1000/10 = 100$ Growth=100* (N-0)/0 = 100*(100-10)/10 = 900%

c) What do our results from parts a) and b) imply about our accuracy in measuring TFP growth?

Explanation: These results tell us that our accuracy in measuring TFP growth depends on our knowledge of the true production function. If we don't know the true function, our measure of TFP growth may be inaccurate.

17. Suppose that city of Willowbee has a production function $Y=AK^{0.5}N^{0.5}$, where A=100 & K=1.

a. Graph this production function with respect to changes in Labour input (show at least 2 accurate points & label all axis & curves). (3)



b. What does the MPN represent intuitively and graphically? (2)

Answer:
MPN is the increase in output (Y) given a 1 unit increase in labour (N).
(That is, the added production from one additional unit of labour).
Graphically, MPN is the slope of the production function at any given level of N.

c. What is the marginal productivity of labour at N=9, assuming A and K are fixed at 100 and 1 respectively. (Note: if it helps, you can say $8^{0.5}$=2.8). (2)

Answer: 17 (or 20)

Calculations: MPN= $\partial Y/\partial N = 0.5AK^{0.5}N^{-0.5} = 0.5*100*1*9^{-0.5} = 50/3 = \sim 17$

(alternative calculation: MPN=$\Delta Y/\Delta N$, at N=9, $Y=100*1*9^{0.5} = 300$. At N=8, $Y=100*1*8^{0.5} = 280$ so MPN=(100-280)/1 = 20)

d. How would MPN be affected by an adverse supply shock? Explain. (2)

Answer: MPN falls

Explanation: As A falls the production function swivels down. This results in a flatter slope (lower MPN) at any N.

Alternative explanation: $MPN=.5AK^{0.5}N^{-0.5}$, so as A falls, MPN falls **(the derivative of MPN with respect to A is positive).**

IV. Labour Market, Employment, and Unemployment

18. Briefly answer the following questions.

a. Explain what is meant by diminishing marginal productivity of a factor?

Explanation:

Initial inputs yield very large increases in production (e.g. first grill and first worker at Sonny's hamburger joint). Subsequent additions of capital and workers still yield production increases, but as more and more are hired, the increases in production drop. The marginal productivity of additional workers becomes smaller (diminishes).

b. What factors (list at least two) cause labour demand to increase?

Factor1: An increase in TFP, or A, will increase MPN, which increases labour demand.

Factor2: An increase in K will increase MPN, which increases labour demand.

c. What factors (list at least two) cause labour supply to decrease?

Factor1: Decrease in population – even if the individuals in the population want to work at the same rate, if the population shrinks, the aggregate supply of labour will fall.

Factor2: Decrease in labour force participation – even if the population remained constant a smaller fraction of this population want to work, so the aggregate supply of workers falls

Other factors that increase labour supply that you might have mentioned are:
*Increase in expected future real wage – want to work more tomorrow & less today, so N^S falls
*Increase in wealth-don't need to work as much to consume same amount, , so N^S falls

19. Suppose that the country of Florin has 100 workers, each with a labour supply function of $N^s=3w$.

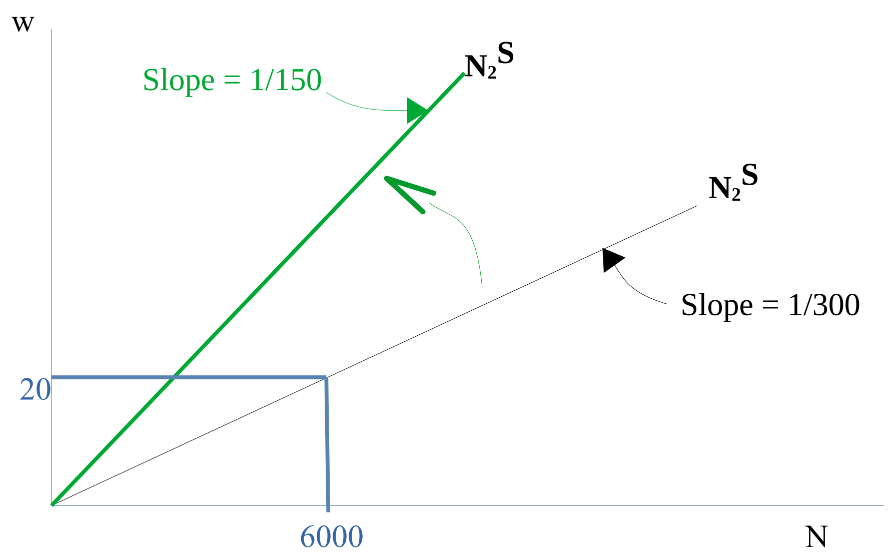
a) Construct and graph the aggregate labour supply curve for Florin. Be sure to label both axis.

Labour Supply curve:

$$N^S_{mkt}=\sum_{1}^{100} N^S=\sum_{1}^{100} 3w=300w \quad \text{-->} \quad w=\frac{1}{300}N^S_{mkt}$$

The answer to question a) is in black. Blue is for part b). Green is for part c)

Labour Market



b. Now suppose the market wage, w* = 20, how much labour will be supplied in Florin?

Answer: 6,000
Calculations: Profit max condition: labour demand is MPN=w where MPN is decreasing
$20 = N*1/300 \rightarrow N=300*20 = 6{,}000$

c. If the government imposes an income tax, t=.5, such that workers only retain 50% of their paid wages, explain and show graphically how the labour supply curve is affected (don't worry about what happens to w*). Derive the new slope of the labour supply curve.

Answer: The labour supply will pivot up (from the origin), with a steeper slope

New slope: 1/150

Explanation: Take-home wage received can be written as (1-t)w = (1-.5)w = (.5)w, so if we substitute this actual take-home wage into the labour supply curve, in place of w, we get $N^S=300(.5)w$, or $N^S=150w$. So the equation for our labour supply line is now: w=1/150 N. Since 1/150 represents a larger number than 1/300, this means the slope of our labour supply curve is now steeper (pivots up from origin).

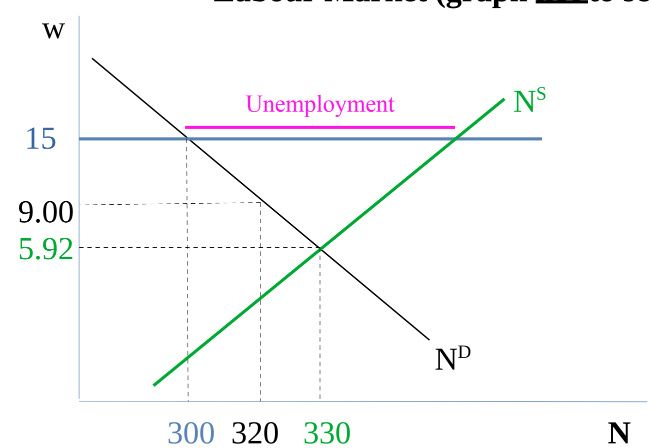
20. Suppose firms' hourly marginal product of labour is given by MPN = A(350 – N), where A=.3

a. How are real wages determined in the labour market? Explain how the MPN function is related to the labour demand function and graph the labour demand curve using at least 3 points.

How Real Wages are Determined: Real wages are determined where labour supply = labour demand in our simple market clearing model	
How MPN is related to labour demand: For the profit maximizing firm in the perfectly competitive market clearing model, labour demand is determined where real wage=MPN, such that the MPN curve represents the labour demand curve, so our curve is w=.3(350-N)= 105-.3N	

The answer to question a) is in black. Blue is for part b). Green is for part c) Pink is for part d)

Labour Market (graph not to scale)



Note: two of the points for part a) are derived in parts b and c, and not duplicated here. The wage at which firms would demand 320 workers is w=105-.3*320 = 9.

b. Let the real wage be fixed at $15/hour. How much labour will the firm want to hire? Show your work.

Labour Demand at $15/hr:	300
Calculation: 15= 105-.3N --> .3N = 105-15 ---> N=90/.3 = 300	

c. Suppose real wages are no longer fixed, but are determined in the labour market equilibrium. Further suppose the aggregate supply of labour is $N^S = 200 + 22w$. Find the equilibrium levels of employment (N) and real wage rate w. Graph the labour market.

W*: 5.92	**N*:** 330

Calculation:

$N^S=N^D$ if $w=105 – .3N^D$, then $N^D=105/.3 – w/.3$

$200+22w = 105/.3 – w/.3$

$.3*200+.3*22w = 105-w$

$(6.6+1)w = 105 – 60$

$7.6w = 45$

$w^* = 5.92$ And at w=5.92, demand is 5.92=105-.3N

so $N^*= (105-5.92)/.3 = 330.27$

d. What would be the impact on the economy of the government fixing the real wage at $15? Explain & illustrate this on your graph.

Impact:	Unemployment.
Explanation: at w=15, labour demand is only 300, whereas labour supply is much higher. The difference between the labour demand and labour supply at w=15 is the amount of unemployment that would be generated by fixing w @15.	

21. Suppose our economy consists of one firm, WeCreateJunk, with MPN=100- ½N.

a) If there are 2 workers in our economy, each with a labour supply function of N_i^S=w, where i=1,2 represents each worker. Solve for the equilibrium labour market wage and employment levels. Show your work!

Equilibrium wage, w*:	50
Equilibrium employment, N*:	100

Calculations: aggregate N^S=w+w=2w, so w= ½N^S

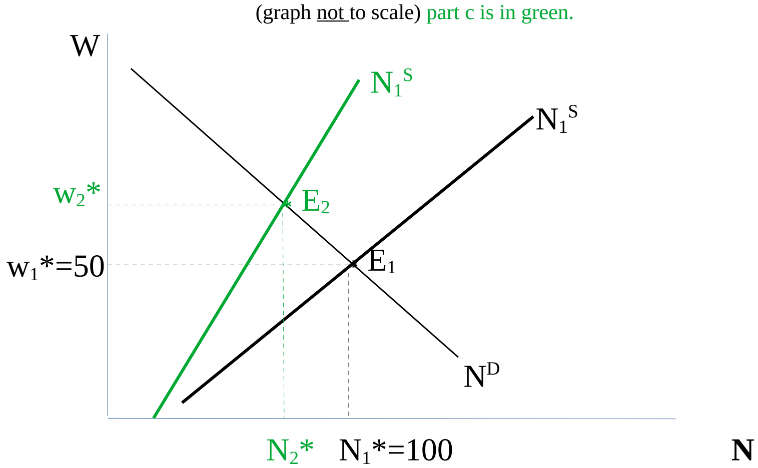
equilibrium → Supply=Demand, so ½ N^S = 100 - ½ N^D
→ N* = 100
→ w* = ½(100) = 50

Alt solution: w=100- ½N so N=200-2w
and equilibrium → 200-2w = 2w → 4w=200 → w=50, N=2*50=100

b) Graph, below, the labour demand and supply functions and label the equilibrium point E_1. (Be sure to label both axis)

Labour Market

(graph not to scale) part c is in green.



c) If an income tax of τ% were imposed on wages (assume τ is between zero and one), would the equilibrium wage and employment be higher or lower? Explain. Show this change using the labour market graph above. Label, but do not calculate the new equilibrium. (2)

New w* higher or lower?	higher
New N* higher or lower?	lower
Explanation: With an income tax on wages, take-home pay is now $(1-\tau)w$ So the labour supply curve would now effectively be $N^S=2(1-\tau)w$, or graphically $w= N^S/(2(1-\tau))$ Since τ is between zero and one, the denominator will be smaller than 2, so the slope of our labour supply curve will be steeper (previously it was 1/2, now it is $1/(2(1-\tau))$ With a steeper labour supply curve, the intersection of Demand and Supply will occur at a higher real wage and a lower level of employment.	
Don't forget to show Show new w* and new N* on graph & label new equilibrium	

d) How would you re-write the new labour supply function to incorporate the tax?

New N^S:	$N^S=2(1-\tau)w$, or graphically $w= N^S/(2(1-\tau))$
Calculations: With an income tax on wages, take-home pay is now $(1-\tau)w$ So the labour supply curve would now effectively be $N^S=2(1-\tau)w$, or graphically $w= N^S/(2(1-\tau))$ Since τ is between zero and one, the denominator will be smaller than 2, so the slope of our labour supply curve will be steeper (previously it was ½, now it is $1/(2(1-\tau))$	

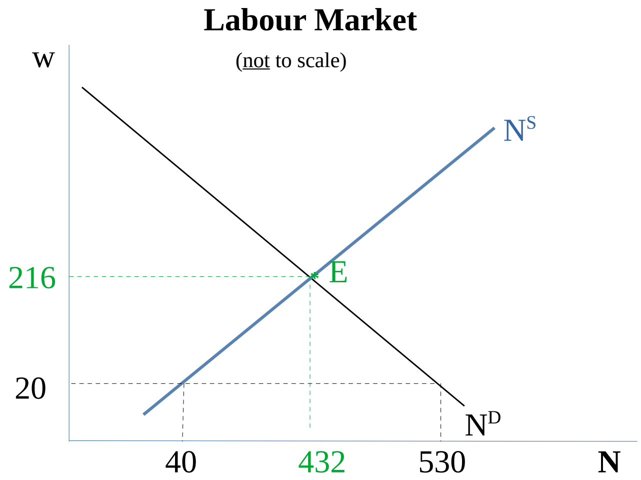
22. Suppose that the country of Mulgore produces hay and has marginal product of labour function MPN=A*540-2N

a. If A=2, what will be the demand for labour in Mulgore when real wage =20?

N^D:	530
Calculations: $20=2*540-2N^D \dashrightarrow 2N^D=2*540-20 \dashrightarrow N^D=530$	

b. Given A=2, graph the demand curve for labour (show at least 2 accurate points).

The answer to question b) is in black. Blue is for part c). Green is for part d)



c. There are three workers (hay bailers) in the economy.
One has a supply function N= ½ w,
another has supply function N=4+ ½ w,
and another has supply function N=-4+w
Calculate the aggregate labour supply (write down the actual function) and graph it, showing at least 2 accurate points.

N^S:	N^S=2w, or graphically w = ½ N^S
Calculations: Aggregate labour supply is N^S= ½ w + 4+ ½ w -4+w = 2w so graphically w = ½ N Two points shown: when w=20, N=2*20= 40, and when N=432, w=432/2 = 216.	

d. Calculate the labour market equilibrium quantity of labour and wage and graph this point (call it E). Be sure to label the equilibrium quantity of labour and wage with their exact values.

N*:	432
w*:	216
Calculations: labour supply is N^S = 2w labour demand is w=2*540-2N or N=540 - ½ w supply = demand --> 2w = 540 - ½w w* =540/2.5 = 216 and N* = 2*216 = 432	

23. The following macroeconomic information is available for the Country of Rivia

	2022	**2023**
Output (books)	9000	8000
Employed	900	800
Unemployed	100	100
Price per Book	9.00	8.00

Calculate the following and be sure to show your work, including the formulas used in the calculation.

a. Growth rate of nominal GDP from 2022 to 2023

N^S:	-11.1%
Calculations:	= 100*[Output(2023)*P_{2023} – Output(2022)*P_{2022}] / Output(2022)*P_{2022} = 100*[8000*8-9000*9]/9000*9 = (65000 - 81000)/81000 = -21%

b. Growth rate of books produced from 2022 to 2023

N^S:	-11.1%
Calculations:	= 100*[Output(2023) – Output(2022)] / Output(2022) = 100*[8000-9000]/9000 = -11.1%

c. Growth rate of real output from 2022 to 2023 in 2022 dollars

Growth in real 2022 dollars:	-11.1%
Calculations:	= 100*[Output(2023)*P_{2022} – Output(2022)*P_{2022}] / Output(2022)*P_{2022} = 100*[8000*9-9000*9]/9000*9 = -11.1%

d. Growth rate of real output from 2022 to 2023 in 2023 dollars

Growth in real 2023 dollars:	-11.1%
Calculations:	= 100*[Output(2023)*P_{2023} – Output(2022)*P_{2023}] / Output(2022)*P_{2023} = 100*[8000*8-9000*8]/9000*8 = -11.1%

e. Why is growth rate the same regardless of which base year you use?

Answer: Because there is only one good, Growth will be the same regardless of base year because there are no shifts of weight from one good to another as prices change (the deflation or drop in price is across all goods -the one good – and doesn't represent a change in relative valuation compared to another product). Nominal growth rate IS, however, lower because it takes into account the reduced overall valuation in the single product produced in the economy.

f. Calculate the unemployment rates for 2022 & 2023, and show your work.

Unemployment rate 2022	-10%
Unemployment rate 2023	-11.1%
Calculations: urate 2022 = 100/(100+900) = 10% urate 2023 = 100/(100+800)= 11.1%	

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24. For each of the following, explain whether the measured amount represents a stock or a flow, and explain:

a) Saving increases by $180 billion in 2020.

Stock or Flow: Flow
Explanation: Saving is what you are not spending out of income over a period of time (2020), it therefore captures a flow of activity for this time window. Therefore, this measured amount is a flow.

b) Canadian merchandise exports fell by 12% in 2020.

Stock or Flow: Flow
Explanation: Merchandise exports records the value of goods you have sold internationally over period of time (2020). Thus the measured amount represents a flow of activity for the time window.

c) Unemployment rose to 1 million people in Ontario by the start of May, 2020.

Stock or Flow: Stock
Explanation: The number of people unemployed measured at a point in time (start of May, 2020), represents a level of stock. Therefore the unemployment level measured at a point in time is a stock.

d) The value of total household wealth in the U.S. had dropped to about 54 trillion by the start of the third quarter of 2008.

Stock or Flow: Stock
Explanation: A measure of wealth at a point in time (the start of Q3-2008) is a measure of stock at a point in time. Therefore, this measured amount represents a stock.

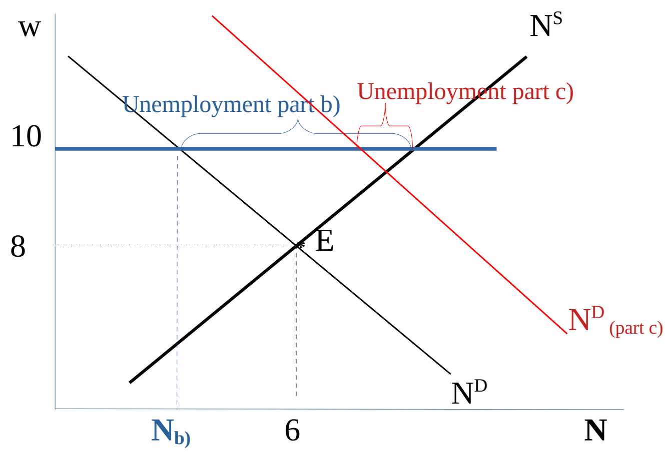
25. Suppose that all firms are identical and all workers are identical, and that aggregate labour demand is $N^D = 10 - ½ w$. Further suppose the aggregate supply of labour is $N^S = 2 + ½ w$.

a) Find the equilibrium levels of employment (N) and real wage rate w. Show your work. Graph the labour market.

N*:	6
w*:	8
Calculations:	labour supply is $N^S = 2 + ½ w$ labour demand is $N^D = 10 - ½ w$ supply = demand --> 2+ ½ w = 10 - ½w w* =8 and N* = 2 + ½*8= 6

part a is in black, part b is in blue, part c is in red

Labour Market (not to scale)



b. Would the level of employment and level of unemployment be higher or lower in this economy if the government fixed the minimum wage at $10? Illustrate this on your graph from part a). Assume that P=1, so that real and nominal wage are the same.

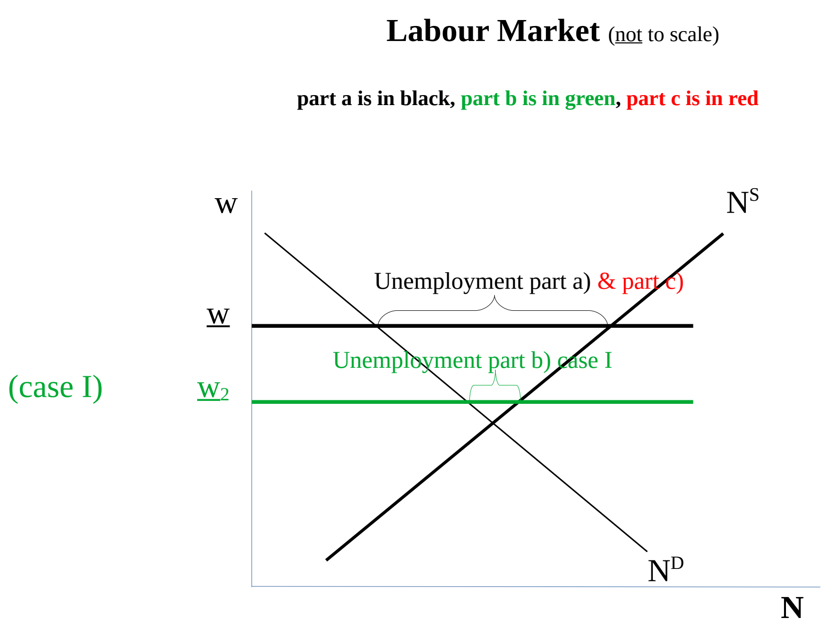
Employment:	Lower, at $N_{b)}$
Unemployment:	Higher, as shown on graph
Explanation: With a minimum wage above w*, employment will be the lowest of either demand or supply, and in this case it is demand. So employment is $N_{b)}$ <N* when w=10. Unemployment is the difference between supply and demand, which at w* & N* is zero, but at w=10 is greater than zero. So Employment has fallen and unemployment has risen.	
Don't forget to show on graph (see blue)	

c. Suppose that minimum wage remains fixed at $10, and price level is unchanged at P=1, but that the economy grows, with more identical firms entering the market. Will this model predict unemployment rises, falls or stays the same? Explain and show graphically using your graph from part a).

Unemployment:	Lower, as shown on graph
Explanation: If minimum wage and price level remain unchanged, such that the real minimum wage remains at 10, but the economy expands with more firms, then the aggregate labour demand curve will increase (rightward), and the level of unemployment will fall (it could disappear altogether if demand increases enough - then the market would clear and unemployment would be zero). The case shown in the graph above shows labour demand rising not quite enough to generate this market clearing outcome.	
Don't forget to show on graph (see red)	

26. Suppose we are in an economy where real minimum wage (nominal minimum wage divided by P) is above above w*

a) Graph the labour market and show the level of unemployment in your graph.



b) Indexing minimum wage to inflation is becoming more common. A province that indexes minimum wage to inflation will increase the minimum wage at the same rate as inflation so that the real minimum wage does not fall. Suppose that the price level rises dramatically from one year to the next. What does the model predict will happen to the unemployment level for a province that does not index their minimum wage (case I), versus a province that does index their minimum wage (case II), relative to part a)?

Unemployment (case I):	The level of unemployment will be lower, as shown on graph
Unemployment (case II)	The unemployment level will not change, it remains the same as part a).
Explanation: In case I (green) minimum wage is not indexed to inflation, so a rise in prices will cause the real minimum wage to fall since $\underline{w}= \underline{W}/P$. This will increase the quantity of labour demanded (see intersection of $\underline{w}_2$ and N^D at higher N) and reduce unemployment. Unemployment could disappear entirely if P rises high enough (and $\underline{w}$ falls far enough), but I have shown a case where some unemployment remains. In case II, because minimum wage rises on par with inflation, $\underline{w}$ is unchanged relative to part a), and therefore unemployment is the same as part a). (see red)	
Don't forget to show on graph	

(Note these two questions are using models which predict unemployment rising with minimum wage, but know that other models can predict the opposite. These questions are NOT intended to be interpreted as a statement of fact about whether minimum wage causes unemployment, it is a statement of what this specific type of models predicts with minimum wage.)

27. Answer the following questions:

a) Define Full-Employment Output

Answer: Full employment output is the amount of output produced at the full employment level of employment – which is the equilibrium level of employment, where $N^S=N^D$ (market clearing framework).

b) Explain what is Frictional Unemployment

Answer: Frictional unemployment is unemployment that arises due to labour market frictions – that is unemployment occurring when workers are briefly unemployed while transitioning between jobs

c) Explain what is Cyclical Unemployment

Answer: Cyclical unemployment is unemployment that arises due to business cycle fluctuations – that is unemployment occurring as the economy rises/falls around full-employment due to recessions/booms causing temporary reductions/increases in jobs

d) Explain what is Structural Unemployment

Answer: Structural unemployment is unemployment that arises due to structural aspects of the worker, or structural changes in the economy – that is poor worker capabilities make a worker less likely to find employment, and/or structural changes in an economy which decrease activity in one sector and increase it in another sector make it difficult for workers in the declining sector to find work. Structural unemployment is usually longer-term unemployment.

e) Which type(s) of unemployment (among the three you discussed above) were most likely to be higher during the height of regional lockdowns in 2020? Explain.

Answer: Cyclical and Frictional would most likely be higher
Explanation: During the regional lockdowns, the economy contracted. This cyclical downturn resulted in lower labour demand for cyclical reasons. Frictional unemployment may also have risen during this time, at least in certain sectors, as previous patterns of job search became disrupted and slowed down.

f) Which type(s) of unemployment (among the three you defined above) were most likely to be higher shortly after regional lockdowns were relaxed in 2021 and the economy was rebounding? Explain.

Answer: Structural would most likely be higher
Explanation: As the lockdowns were relaxed, cyclical and frictional unemployment would be declining because the economy began to expand again and methods of job search improving, but Structural unemployment would be increasing because there were shifts to our economic structures resulting from the pandemic & pandemic measures (e.g. greater demand for on-line purchasing, delivery service, medical equipment manufacturing). Thus, we would expect to see shorter term unemployment rates falling, while long term unemployment rates rose.

g) Would you expect the official unemployment rate to be a good measure to track changes labour market outcomes as firms shifted to posting more part time and contract jobs? Explain.

Answer: No, I would not expect it to be a good measure.
Explanation: Part-time and contract work are frequently not utilizing labour to it’s desired capacity. Therefore, the official unemployment rate wouldn’t be the best measure to capture this decline in labour outcomes as it fails to capture under-employment.

28. Suppose that you are using a Search and Matching Framework to understand unemployment levels in the economy. Assume that you have designed a general matching function M(U,V) which predicts the number of matches formed increases with the number of workers searching and the number of job vacancies posted, there is free entry into and out of the market, and you assume a fixed job destruction rate of δ (which is a fraction between zero and 1). Also suppose that you assume a fixed labour force (LF) where the level of unemployment is then defined as U=LF-E, where E is the number employed.

a) What condition do you use to calculate the steady state level of employment and unemployment in the economy? Explain.

Answer: The condition you use is Flows Into E = Flows Out of E
Explanation: When you set Flows in equals flows out of employment, you can calculate a steady state amount of Employment, E*, that will exist when the economy is settled in a steady position. The steady state level of employment can then be calculated: U* = LF - E*

b) If there is an increase in the labour force (due to an increased participation rate, say), will your model predict the steady state level of employment to rise, fall, or stay the same? Explain.

Answer: The steady state level of employment should rise
Explanation: An increase in the labour force means that more people will be searching for work (as these additional people first enter the labour force, they don't start out with jobs, they start out as unemployed). Greater numbers of people searching means that greater numbers of matches will be forming. This increases the value for firms to post vacancies (it will rise above zero) so more firms will enter the market (free entry) and post jobs. As the matches continue to form, Employment will rise. Jobs will still be destroyed in this economy, but given that the job destruction rate does not exceed 1, a positive proportion of the increased matches will stay matched, meaning steady state employment level, E*, will rise.

c) If match productivity (output from a match) rises, and the labour force remains constant, will your model predict a rise/fall or no change in the steady state level of unemployment? Explain.

Answer: The steady state level of unemployment should fall.
Explanation: An increase in the productivity of a match means that the value of a match rises, so the value to firms of posting a job rises above zero. As there is profit to be made from posting a job, more firms enter the market (free entry) and post jobs. The match rate rises, and E* rises. Because U*=LF-E*, and E* is rising while LF is constant, then U* will fall.

d) If the job destruction rate in the economy were to rise, will your model predict a rise/fall or no change in the steady state level of employment? Explain.

Answer: The steady state level of employment should fall
Explanation: An increase in the job destruction rate means that of those that are currently employed, they will lose their jobs at a greater rate. This will decrease the numbers employed. Although a decrease in the numbers Employed will decrease the numbers of jobs destroyed, the rate of destruction of jobs (employed) remains higher than it was before. Moreover, with higher job destruction rates, the value of posting a job falls, so as jobs are destroyed firms will begin to leave the market and the match rate falls. With both greater job destruction and lower match rates, the steady state level of employment, E*, should fall. (There are some unrealistic scenarios where this wouldn't occur, e.g. if we started at E*=0, it wouldn't fall, but for any realistic scenario, E* would fall)

e) For parts b), c) and d) will the change happen instantly or gradually over time? Explain (be sure to refer to changes in flows).

Answer: Gradually
Explanation: In cases b & c, the impact of the change is to increase the flow into Employment due to increased matches formed. In case b, the increase in the labour force increases U (search models typically don't assume people enter the LF as employed already), increasing U increases value of a match & match rates, which gradually increases E. In case c, the increase in the productivity of a match increases the firms entering the market (increases V), increasing the match rates, which gradually increases E (and therefore gradually decreases U). In case d, the increase in job destruction is a rate of destruction. So people won't just instantly lose their jobs. They lose these jobs at a slightly higher rate as time passes, and the matching rate also falls gradually as jobs are destroyed and firms choose not to post again.

f) If you had a set of algebraic equations to fully specify this model, would you feel more comfortable or less comfortable talking about the model's predictions? Explain.

Answer: There is no correct answer for this, I'm just curious about your perceptions.
Explanation: My own personal comfort level is increased with the equations because then I can check to make sure I haven't missed anything (my thinking is more disciplined). Of course, just because I feel more comfortable, doesn't mean I'm less likely to miss things!

<u>V. Consumption, Saving and Investment</u> (for this entire section, assume a closed economy)

29. Answer the following questions:

a) Explain why Prudence may wish to smooth her consumption over time, and how Prudence might go about doing so.

Explanation of Why & of How: Why – Prudence may wish to smooth her consumption so that she is able to maintain a roughly consistent quality of life over time. Importantly, Prudence would not want to consume everything today and starve tomorrow, or vice versa. How – Prudence can smooth her consumption by borrowing (or dissaving) when her income is low, and saving during that part of her life where her income is high.

b) Explain the why the substitution effect might cause individuals to decrease current period consumption when the real interest rate rises. Then explain the income effect for borrowers when said interest rate rises. Do these work in the same or opposite directions?

Substitution Effect Explanation: When r rises, cost of current period consumption relative to future consumption increases. Thus, individuals would *substitute* away from (decrease) current consumption.
Income Effect for Borrowers Explanation: For borrowers, higher r means lower lifetime income (greater costs of borrowing), with lower lifetime income borrowers will need to reduce consumption and will reduce both current and future consumption in order to smooth consumption over time.
Same or Opposite direction: Same direction. For a borrower, the income effect on C is negative too.

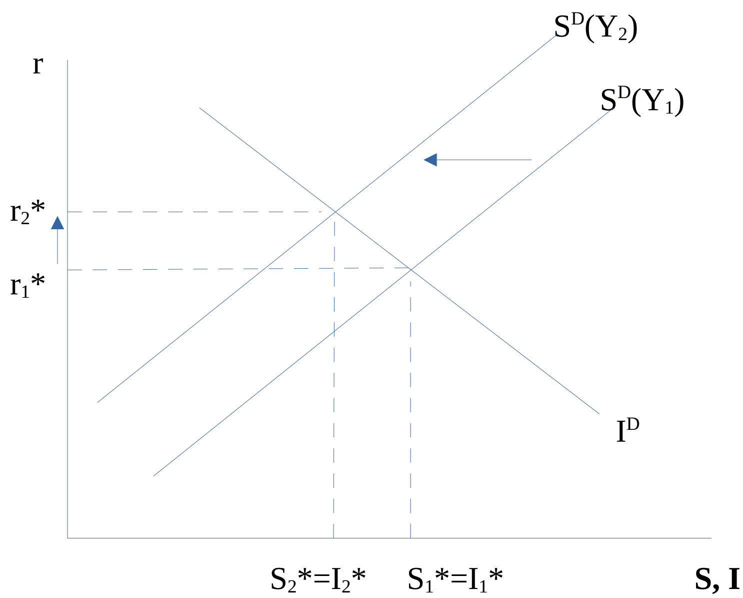
c) Of the following cases, explain which are consumption smoothing behaviours and which are not.

- **i)** Real interest rates rise and Prudence (who has interest deferred student loans decreases her current consumption slightly and expects to reduce her future consumption slightly as well.
- **ii)** Prudence receives a bonus of $50 today and decides to spend it on a $50 bottle of Canadian Whisky to share with her friends tonight.
- **iii)** Prudence expects a raise of $10,000 next year, so she decreases her savings slightly in the current period to purchase slightly more groceries.

Answer: i and iii are decisions that are derived from consumption smoothing, ii is not.
Explanation: In case i, an increase in r will decreasePrudence’s available lifetime income (because she is a borrower and will need to pay this back in the future). Prudence doesn’t need to pay student loans back right now (interest deferred), but she will need to pay it back in the future. However, rather than solely reducing her future consumption, Prudence has also reduced her current consumption. Therefore Prudence is smoothing the negative impact on consumption over time. In case iii, Prudence’s lifetime income has risen, and so she has more money with which to consume over her lifetime. Because she increases her current consumption now, she is smoothing her consumption by consuming some of that increase in the current period. In case ii, however, Prudence decides to spend her entire bonus in the current period, so she is not smoothing her consumption.

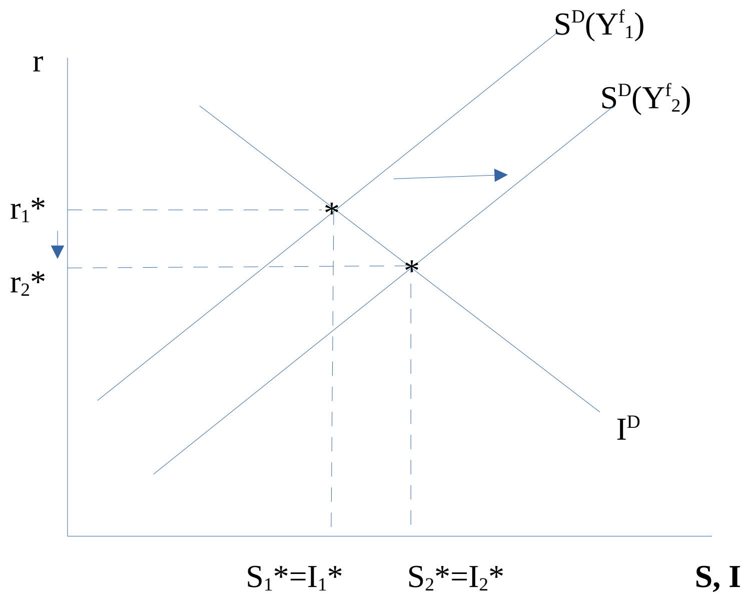
30. Explain and graph the impact on the saving demand curve and the equilibrium r* of the following changes:

a) Current Y falls because everyone has caught this flu and is less productive!



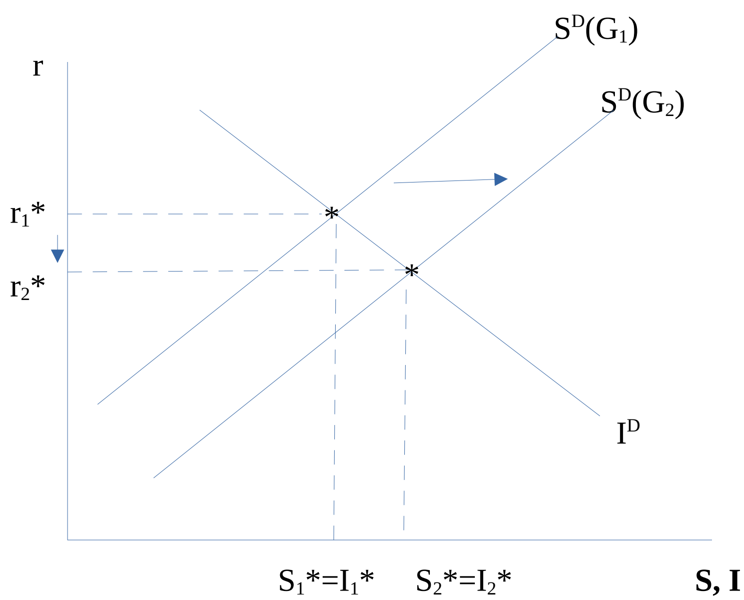
Impact on S^D: Falls (shifts left)	
Impact on r*: rises	
Explanation: A decrease in current Y will decrease C^D, but not by full amount (generally by some marginal propensity to consume) and since $S^D = Y-C^D-G$, where Y is falling by more than C^D, then S^D will decrease. Notice equilibrium r* rises because the intersection of S^D and I^D occurs at a higher r.	

b) Expected future Y falls.



Impact on S^D: rises (shifts right)	
Impact on r*: falls	
Explanation: Decrease in Y^f will want to smooth decreased lifetime income over time, so will decrease current C^D, thereby increasing S^D since $S^D = Y - C^D - G$, and current Y is constant while current C^D is falling. Notice equilibrium r* falls because S=I at lower r.	

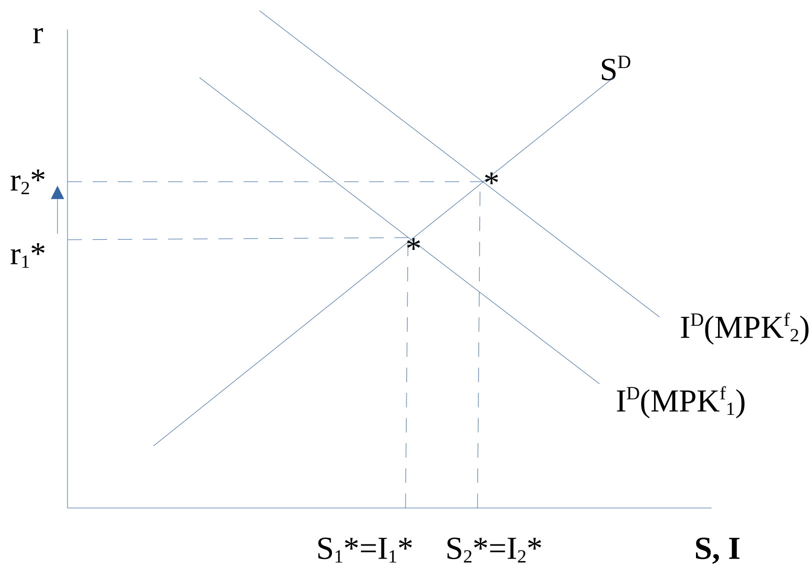
c) A new government is elected that wants to balance the budget by cutting spending.



Impact on S^D: increases (shifts right)	
Impact on r*: falls	
Explanation: Government cuts spending so G falls --> increasing S^D since $S^D = Y - C^D - G$ and the negative component is shrinking so the total rises. Notice equilibrium r* falls because S=I at lower r.	

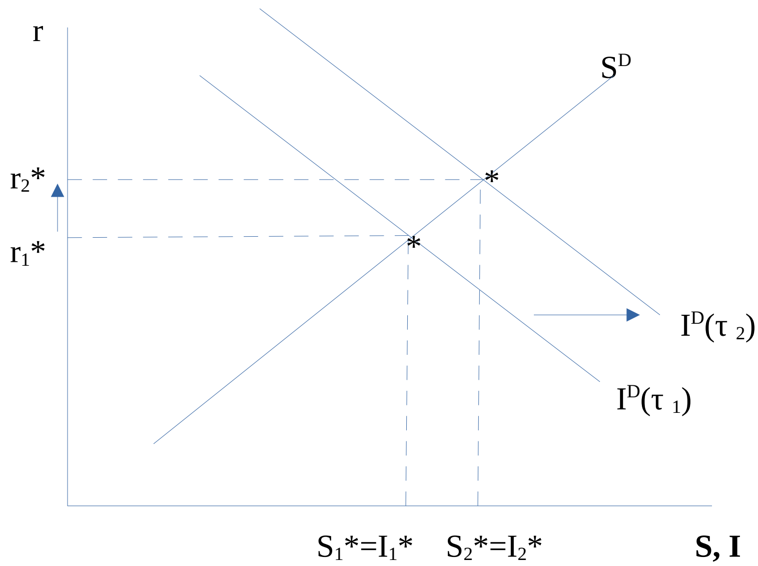
31. Explain and graph the impact on the investment demand curve, equilibrium r, S and I, of the following changes:

a) A technological breakthrough increases MPK^f



Impact on I^D: Increases	
Impact on r*: Increases	
Impact on S* & I*: Increases	
Explanation: An increase in MPK^f will increase K*, and since $I^D = K^* - K_t + dK_t$, an increase in K* will increase I^D –the curve will shift rightward. The point at which S^D=I^D occurs at a higher r, and both S* and I* will rise.	

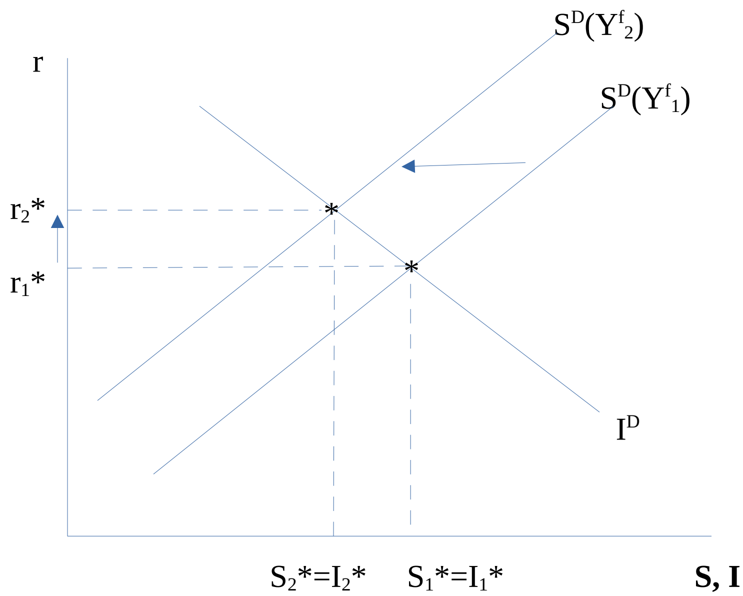
b) Effective tax rate on capital, τ, decreases.



Impact on I^D: Increases	
Impact on r*: Increases	
Impact on S* & I*: Increases	
Explanation: If τ falls, you can think of the impact one of two ways: 1. the tax adjusted MPK^f rises (pivots up) or the tax adjusted user cost shifts down (parallel). In either case, K* and since $I^D = K^* - K_t + dK_t$, an increase in K* will increase I^D –the curve will shift rightward. The point at which $S^D=I^D$ occurs at a higher r, and both S* and I* will rise.	

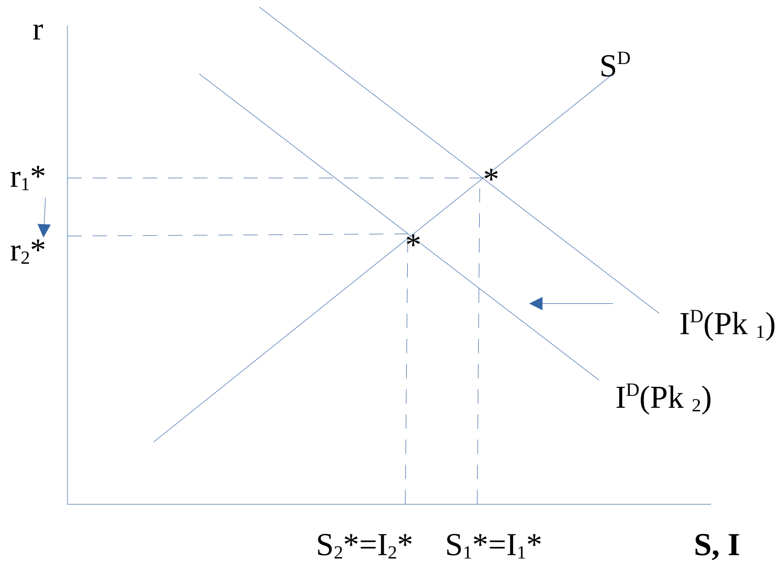
32. Explain and graph the impact on the goods market (curves and equilibria r*, S*, I*) of the following changes:

a) Increase in Y^f.



Impact on I^D: unchanged	
Impact on S^D: falls (shifts left)	
Impact on r*: rises	
Impact on S* & I*: lower S* & lower I*	
Explanation: Increase in Y^f will want to smooth decreased lifetime income over time, so will increase current C^D, thereby decreasing S^D since $S^D = Y-C^D-G$, and current Y is constant while current C^D is higher. Notice equilibrium r* rises because S=I at higher r. S* and I* are occurring at a lower level.	

b) Temporary rise in MPK and Pk simultaneously.



Impact on I^D: lower
Impact on S^D: lower
Impact on r*: lower
Impact on S* & I*: lower S* and lower I*
Explanation: . Because the increase in MPK is temporary, then MPK^f does not increase, so this MPK change will have no impact on K*. However, the increase in Pk will increase UC, causing K* to fall. Since $I^D = K^* - K_t + dK_t$, an decrease in K* will decrease I^D –the curve will shift leftward. Since I^D falls, the intersection of I and S occurs at a lower r, and lower S* and I*. Note that it doesn't matter if the effect on Pk is permanent because Pk enters contemporaneously into the K* and I^D decisions. *If MPK^f were to also increase, then we'd have two opposite effects – a rise in MPK^f would increase K*, while the increase in Pk increasing UC would decrease K*. So the effect on I^D, and therefore everything else (since I^D is the only shift) would be ambiguous.*

33. Consider a closed economy and the following output, desired consumption and desired investment:

$Y = 1000$ (full employment output)

$C^D = 300 + 0.3Y - 175r$

$I^D = 500 - 250r$

$G=0$

a) Find the real interest rate that clears the goods market and show your equation for desired savings.

Equation for S^D: $S^D = 400 + 175r$
r*: 0.24
Explanation: $S^D = Y - C^D - G$ $= Y - 300 - 0.3Y + 175r - 0$ $= 0.7(1000) - 300 + 175r$ $= 400 + 175r$ in Equlibrium: Sd=Id so $400 + 175r = 500 - 250r$ $r^*=100/425 = 0.24$

b) Suppose now, we introduce G = 100. What is the effect on r* and how is the desired savings equation changed?

Equation for S^D: $S^D = 300 + 175\ r$
r*: r* rises to 0.47
Explanation: $S^D = Y - C^D - G$ $= 300 + 175\ r$ using equilibrium condition $S^D = I^D$, we arrive at: $300 + 175r = 500 - 250r$ so $r^* = 0.47$ Savings equation decreased for any given level of Y and C. This implies the savings curve decreased (shifted left), so the interest rate that clears the good market increases at any given level of Y.

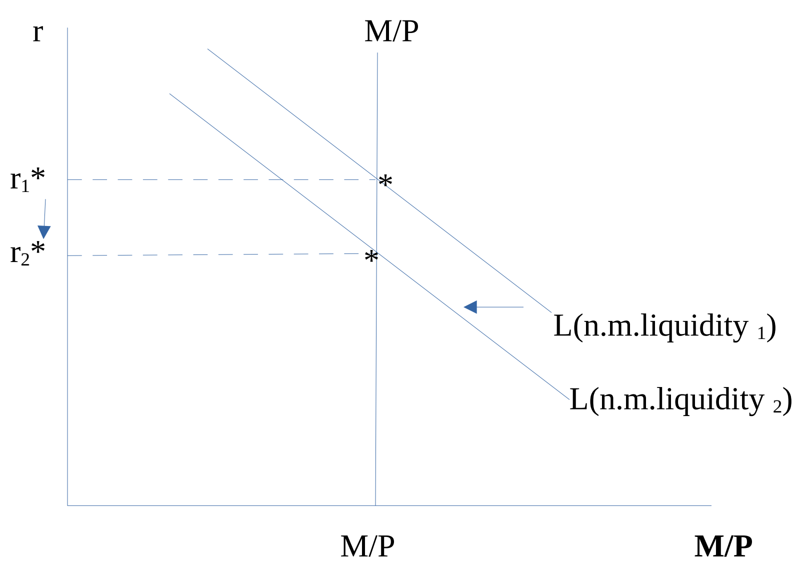
VII. Money and Inflation

34. Suppose that the country of Minionland is deciding whether to use Bananas and Beads as their currency. Assuming that either would be an acceptable form of payment for goods and services, explain which of Bananas or Beads would be the more suitable choice for their currency.

Bananas or Beads more suitable currency: Beads
Explanation: Both can be the currency because both would be accepted as payment. However, beads achieve more of the functions of currency than bananas in that beads would make a better medium of exchange being lighter and easier to transport, better able to store value over time because they do not rot like bananas. Both bananas and beads would be semi-useful in terms of unit of account. Aside: the benefit/drawback to both (but especially bananas) is that they tend to consume it in and of itself so it would be beneficial in that it could serve as both a medium of exchange and a consumption item, but detrimental because it would be hard for them to control their consumption if they always had the consumption item on hand.

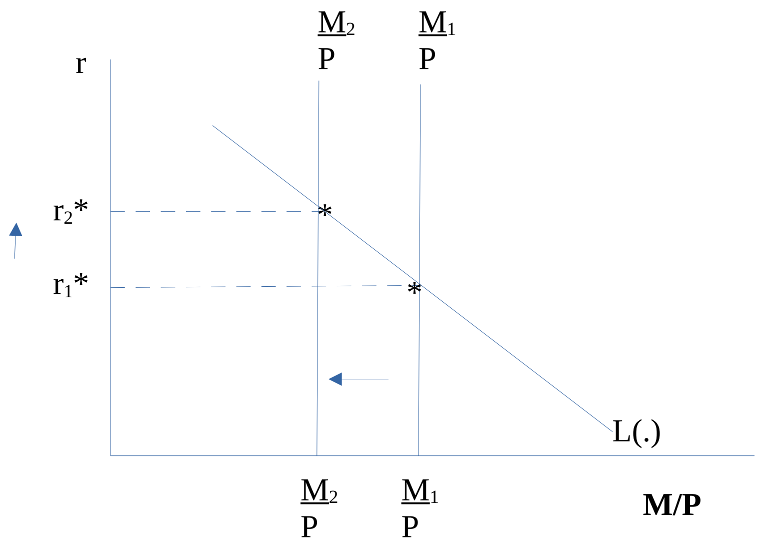
35. Show graphically and explain the impact of the following on the money demand and/or money supply curves, and the equilibrium real interest rate in the Asset Market.

a) Increase in liquidity of stocks & bonds, impact on L(.) and r*



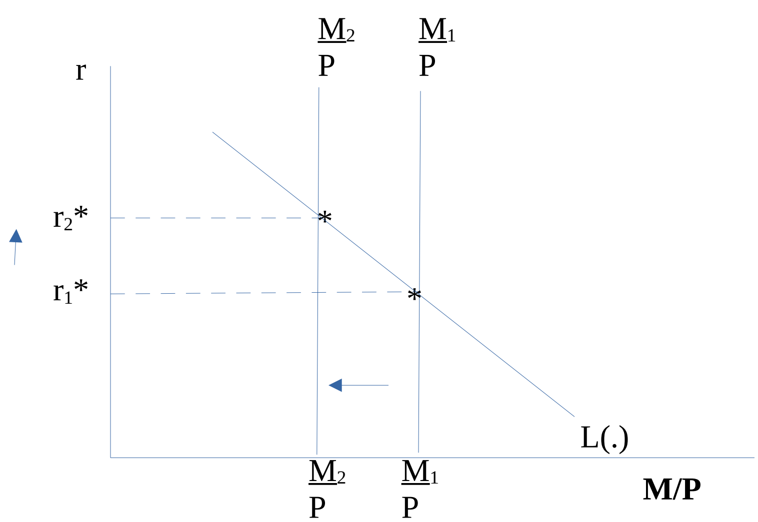
Impact on L(.) and or M/P: M/P is unchanged, L(.) falls (shifts left)
Impact on r*: falls
Explanation: .Liquidity of stocks and bonds goes up, less need to hold money, money demand decreases – L(.) shifts left. The new r* that clears the asset market is lower (M/P = L(.) at lower r*)

b) Contractionary monetary policy (to try to curb inflation) causes Ms to fall --show graphically, & what is impact on r*



Impact on L(.) and or M/P: M/P falls (shifts left), L(.) unchanged
Impact on r*: rises
Explanation: . A contractionary monetary policy which causes MS to fall results in a leftward shift of M/P. The r* that clears the asset market (where L(.)=M/P) rises.

c) People begin to distrust banks and the Currency/Deposit ratio of public rises.



Impact on L(.) and or M/P: L(.) unchanged, M/P shifts left (increases)
Impact on r*: increase
Explanation: A rise in cu will increase the numerator and denominator of the money multiplier (mm=cu+1 / cu + res) by the same amount, but because the denominator is smaller than the numerator, the % increase of the denominator will be larger, therefore the mm will shrink, and so will M^S, since M=mm*Base. A drop in money supply will shift M/P left, increasing the r* that clears asset market.

36. Answer the following questions

a) Economists often refer to tradeoffs in decision-making, which tradeoff do Okun's Law and the Phillip's curve model?

Answer: The trade off between unemployment and inflation
Explanation: .The Phillips curve models a negative relationship between unemployment and inflation, while Okun's law may be used to help quantify that relationship by noting that a 1% departure from FE employment (unemployment change) results in a 2% departure from FE output (which will impact prices). Generally a slope in the range of -2 is a common estimate for the slope of the Phillips curve. There is some debate as to whether this relationship holds in recent years. It appeared to be the case in the 1960s, but in the 1970s with stagflation, the relationship was positive. A modified version of the Phillips curve, one that took inflation expectations into account, could be used to predict that positive relationship. But this may be too contrived.

b) Explain why the Bank of Canada uses and continues to use a 2% inflation target?

Explanation: The Bank of Canada continues to use a 2% inflation target because maintaining consistency (three plus decades targeting 2%) is thought to increase credibility and stability. Historically and internationally, countries have used targets in a nearby range, 1% being another common one, although some countries target higher. Because targets may be imperfectly met, having a target above zero helps to avoid over-slowing the economy. Having a low target, like 2% can avoid high inflation and it's associated costs.

c) Describe three real-world examples of the costs of inflation

Answer: e.g. interwar Germany *Demand to be paid more frequently *purchasing groceries daily to avoid inflated prices by waiting e.g. present day Argentina *purchasing items that may not be needed now in order to buy before prices rise *lack of confidence in own currency *shop owners have a hard time determining what price to set for goods You may also reference increased risk, and decreased ability to collect tax and provide public goods, with both individual (personal) and national examples.

d) Using what you have learned about the costs of unemployment, explain how unemployment has both individual and aggregate impacts.

Answer: Aggregate impact of unemployment is less than full employment level of output. Individual impact of unemployment is temporary loss of income, which could also have impact on other family members, and friends, and can have long run impact if, for example, this loss of income means inability to support a secondary business, mortgage or other loan.

e) Are the costs of inflation and unemployment temporary or permanent or both? Explain.

Answer:
These costs can be both temporary and permanent. Temporary impact of unemployment is loss of current income (aggregate and individual). Temporary impact of inflation is difficulty purchasing items in the present. Longer run impacts can include loss of business or other assets for which one cannot make loan payments. As well as potential longer term growth implications because of these losses.

37. Consider how our fractional reserve banking system influences money supply via the multiple expansions of loans and deposits.

a) If the public decides to keep more cash in pocket, will this decision increase or reduce the effect of the fractional reserve system on the expansion of money supply? Explain intuitively and show algebraically.

Answer: Reduce.
Explanation: Intuitively, the more money kept in pocket is less held in banks. So it will reduce the amount of reserves available to expand the money supply via the fractional reserve system.

Algebraically, the money multiplier = (cu+1)/(cu+res).
Because res<1, an increase in cu will increase the denominator proportionally more than the numerator, thereby decreasing the multiplier, and decreasing the expansion of money supply.

b) What are the implications of a higher reserve deposit ratio on the money supply? Explain.
Answer: Lower money supply
Explanation: A higher reserve deposit ratio means the banks are keeping more money in reserve (lending less out, and thereby reducing the amount of expansion occurring via fractional reserve system).

Algebraically, the money multiplier = (cu+1)/(cu+res).
An increase in res causes the denominator to rise, and decreases the expansion of money supply.

c) From a bank's perspective, list one advantage and one disadvantage of a low reserve deposit ratio.
Advantage:
A lower reserve deposit ratio means more loans on which they earn interest.

d) What would happen to money supply if the reserve deposit ratio were increased to 1 (100%)?
Answer: Money supply would equal the Base
Explanation: If the reserve deposit ratio were increased to 1, the banks would not be expanding the money supply by the multiple expansion of loans and deposits, as they would not be lending out any fraction of the deposits! Algebraically, the money multiplier = (cu+1)/(cu+res). If res=1, the numerator equals the denominator, so the multiplier=1, which means the money supply simply equals the Base - - it is not multiplied beyond that.

38. What is the asset market equilibrium condition?

Answer: Money Supply = Money Demand

39. Suppose that in the country of Vanyassa the public decides to keep all currency in pocket and not use banks. What is the money supply in this economy?

Answer: Money Supply = Base in this economy
Explanation: There is no expansion of loans and deposits in this economy because no one uses commercial banks. Therefore the money supply in the economy equals the base.

40. Answer the following questions:

a) Define the money multiplier
Money Multiplier: is the dollars of money supply that can be created per base dollar

b) What is the value of the money multiplier in a system of 100% reserve banking, that is res=1?
Answer: Money Multiplier = 1
Explanation: money multiplier = (cu+1)/(cu+res) = (cu+1)/(cu+1) = 1

c) Given: M=1,000,000, CU=400,000, and res=0.3, what is the value of the money multiplier, bank reserves, BASE, and deposits?

Answer: Deposits = 600,000
Explanation: M=CU+DEP, so DEP=M-CU = 1mil – 400K = 600K

Answer: Reserves = 180,000
Explanation: res = 0.3 = Reserves/Deposits.
So 0.3=Reserves/600K --> reserves = .3 *600K= 180,000

Answer: Money Multiplier = 1.72
Explanation: money multiplier = (cu+1)/(cu+res), where cu=CU/DEP = 400K/600K = .66667
so money multiplier = (.67+1)/(.67+.3) = 1.724

Answer: BASE = 580,000
Explanation: M/Base = money multiplier, so BASE=M/mm
if M=1mil and money multiplier = 1.72, then BASE = 1mil/1.724 ~= 580,000

d) Suppose there are three countries (A, B, and C). Each country has a BASE of 1 million. Which country will have the greatest money supply – Country A, which has no banks, and only currency, Country B which has banks, where people hold deposits, but the reserve deposit ratio is 100%, or Country C where there are banks and people hold deposits, and the reserve deposit ratio is 50%?

Answer: Country C
Explanation: The money supply will equal the base in both A and B, but in C, the fractional reserve system will expand the money supply to something larger than 1 million.

41. Describe the effect of each of the following on M:

a) An increase in muggings causes people to deposit more of their cash in
their bank accounts.

Answer: Increase M
Explanation: Less cash in pockets means more expanded by banks. Algebraically, the money multiplier = (cu+1)/(cu+res). Because res<1, a decrease in cu will decrease the denominator proportionally more than the numerator, thereby increasing the multiplier, and increasing the expansion of money supply.

b) Increased stability causes banks reduce their reserve deposit ratio.

Answer: Increase M
Explanation: Lower reserves means greater money expanded by banks. Algebraically, the money multiplier = (cu+1)/(cu+res). A drop in res will reduce the denominator -->increase the multiplier --> increase money supply.

c) The government buys government bonds from the Bank of Canada with newly printed currency

Answer: Increase M
Explanation: Increase M because M= BASE* money multiplier. Purchasing bonds with newly printed money will increase the BASE and thereby increase money supply.

VI. Trends, Cycles and Inequality

43. Answer each of the following questions:

a) Explain (in detail) why it is difficult to determine whether real wage is cyclical or not.

Explanation: If you are using aggregate data on wages – wage is typically only recorded for those who are working. The people who lose their jobs may be at the upper or lower end of the distribution, depending on what causes the economy to contract, so the average wage may increase or decrease depending. People who lose their jobs are not recorded as zero wages, they are simply taken out of the averages. So some contractions may have wages that are procyclical, whereas others may be countercyclical –depending on which segment loses the jobs. If it is the lower earners, wages would actually be countercyclical. If you are using longitudinal data, workers wages are tracked across time, and loss of income can be tracked not only in terms of averages, but also in terms of individual declines and even measure complete loss of income. It is expected that longitudinal data may be better for picking up procyclical wage patterns because you can track individual wages relative to what they were prior to the economic contraction, and not confound wage changes with employment composition changes.

b) Are durable consumption good more or less pro-cyclical than non-durable consumption goods? Explain.

Answer: More pro-cyclical
Explanation: In the case of non-durable consumption goods like food, we don't have as much leeway to reduce it when the economy contracts (we gotta eat!). Whereas purchases of durable consumption goods (cars, appliances) can be postponed or even foregone entirely. For example, in an economic contraction, when individuals lose their jobs, they would reserve the money they do have for food, and forego buying a new dishwasher until earning again.

44. Answer the following questions on business cycles:

a) How are business cycles measured?

Answer: Peak to Peak or Trough to Trough

b) Why is it difficult to define recessions

Answer: Because there is no clear consensus as to how long a contraction must last, or how deep the impact, or how widespread the contraction needs to be in order to be classified a recession. Moreover, each business cycle will vary along these dimensions –some lasting longer but having a smaller contraction, or being less widespread – but each type of recession may severely damage the economy in both the short and long term.

c) list 3 ways you might quantify a recession.

Answer: Three major areas of quantification are depth, length and breadth measures.
1. By the depth of the recession (e.g. how far AEA falls)
2. By the length of the recession (how long the economy is contracting)
3. By the breadth of the recession (how many sectors/components of the economy are affected).

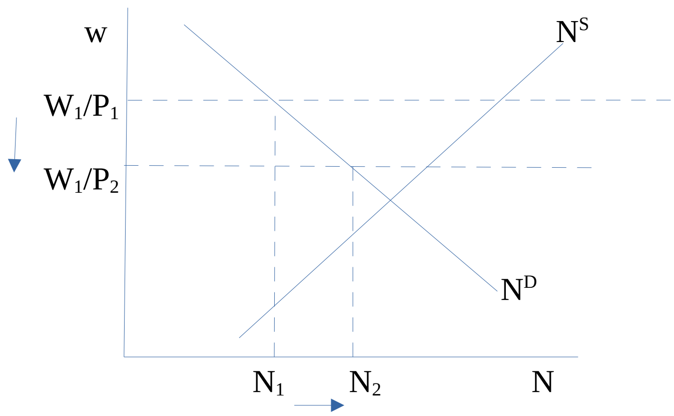
d) If you had to choose between RBC and sticky wage approaches, which would be better equipped to try to understand the labour market impact of business cycle fluctuations? Explain.

Answer: Sticky wage approach
Explanation: Basic RBC models are market clearing models, therefore they will not predict fluctuations in unemployment during the business cycle. But rises in unemployment during economic downturns are are a major focus of concern with the labour market impact of business cycle fluctuations. Sticky wage models, while they may not be the best labour market model to use (and won't always predict unemployment), are better equipped than the RBC models to understand the implications of business cycles on unemployment.

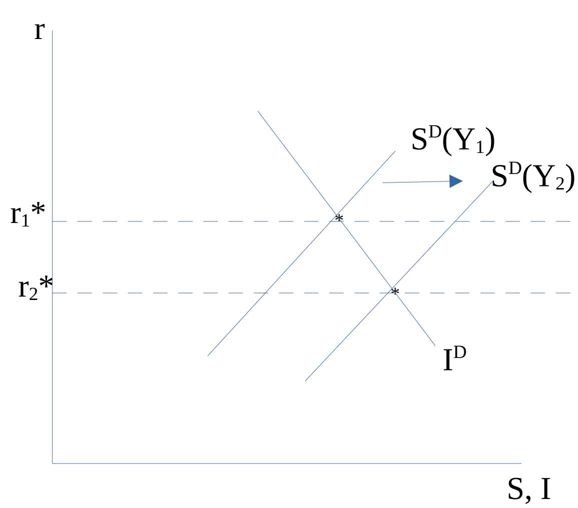
45. Suppose that the Bank of Canada implements policy that increases money supply. Assume some price movement but complete wage rigidity in this time frame (short run), and assume the initial W/P > w*, to answer the following questions.

a) Show the impact of this policy in the Asset Market, Goods Market and Labour Market.

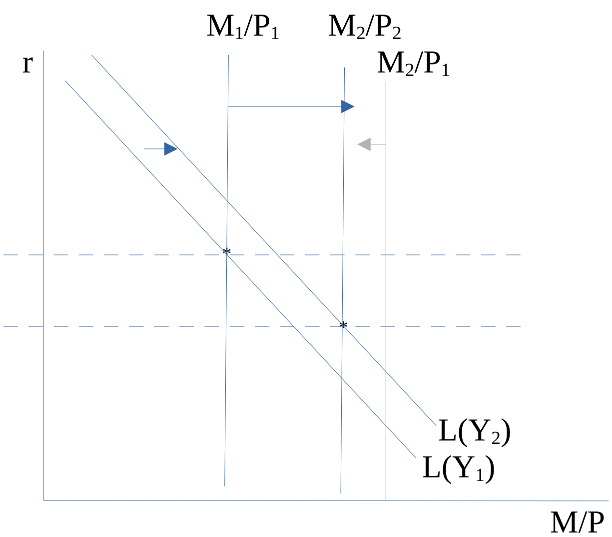
Labour Market



Goods Market



Asset Market



Description: A policy that expands money supply will shift M/P to the right, reducing r*. With lower r, Consumption will increase and thereby aggregate demand ($Y=C^D+I^D+G$) will increase. With a boost in Y, money demand will increase (shift right), and Saving demand will increase (shift right). The increase in aggregate demand puts upward pressure on price, as P rises (M/P shifts back left a bit, but not so much as to supersede the original impact), W/P falls and employment increases (assuming we started out with W/P stuck above the market clearing wage.

You may note that I* also rises, due to a drop in r* (a movement along the I^D curve). It will not reverse the positive consumption -AD effect.

b) In the short run, when wages are sticky, explain what happens to Employment, Saving, real interest rates, GDP and prices.

N: rises
S: rises
r: falls
Y: rises
P: rises
Explanation: Same as description above.
A policy that expands money supply will shift M/P to the right, reducing r*. With lower r, Consumption will increase and thereby aggregate demand will increase. With a boost in Y, money demand will increase (shift right), and Saving demand will increase (shift right). The increase in aggregate demand puts upward pressure on price, as P rises (M/P shifts back left a bit, but not enough to increase r back to r_1*), W/P falls and employment increases (assuming we started out with W/P stuck above the market clearing wage.

46. Answer the following questions related to Business Cycle Transmission (uses some open economy concepts):

a) Your friend is concerned about a recession in the UK having implications on the Canadian Economy. Explain the likely impact on the Canadian economy of a recession in the UK.

Explanation:
A recession in the UK would reduce the UK demand for Canadian Exports. This would reduce Canada's NX and therefore reduce Y=C+I+G+NX in Canada –having a recessionary impact in Canada. However, the impact would be small if Canada exports a relatively small proportion of it's Exports to the UK.

b) Your friend has now started worrying about the potential decline in growth of China's economy. Explain the likely impact on the Canadian economy of economic contraction China.

Explanation:
Slower Growth in China implies GDP is not growing as fast as it was before. This decline means a decline in growth of Y, so Imports in China will be growing slower, which means NX in Canada will be growing slower. A reduced growth in NX means a reduced growth in Y for Canada (Y=C+I+G+NX) in Canada –having a recessionary impact in Canada. However, the impact will depend on how much Canada typically exports to China (currently tens of billions of dollars in trade).

47. Your friend pays attention to many economic indicators, and has predicted an imminent recession due to a decline in the average work week (hours) in Manufacturing.

a) What concerns might you raise about relying on this variable to predict recessions? You may wish to use FRED data to support your statements.

Concerns:
Like other leading indicators, the average work week is not consistent in it's relationship with recessions. Sometimes it decreases just before a recession, but sometimes it decreases well before a recession, and sometimes it decreases after a recession starts. Moreover, sometimes it decreases and no recession follows. https://fred.stlouisfed.org/series/AWHMAN

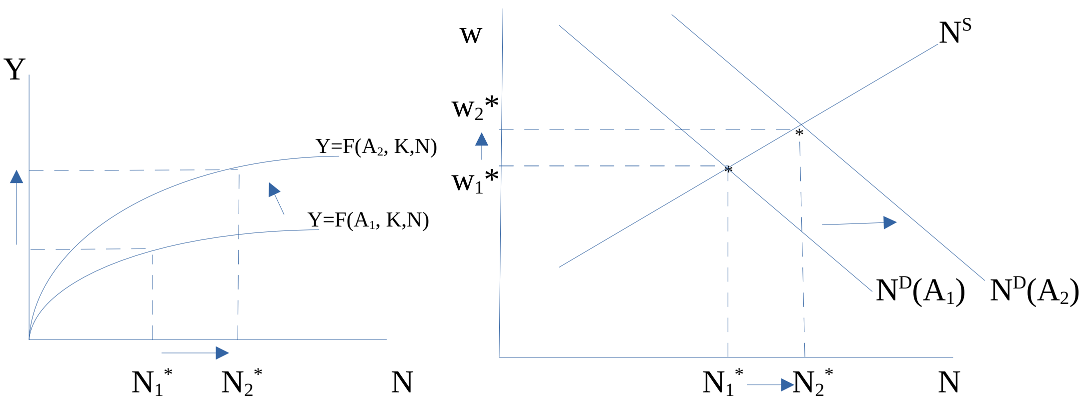
Moreover, if a country is shifting away from Manufacturing as the main driver of aggregate economic activity, then this variable may not be as significant a predictors of future business cycles.

b) List two reasons why it is difficult to predict recessions, in general.

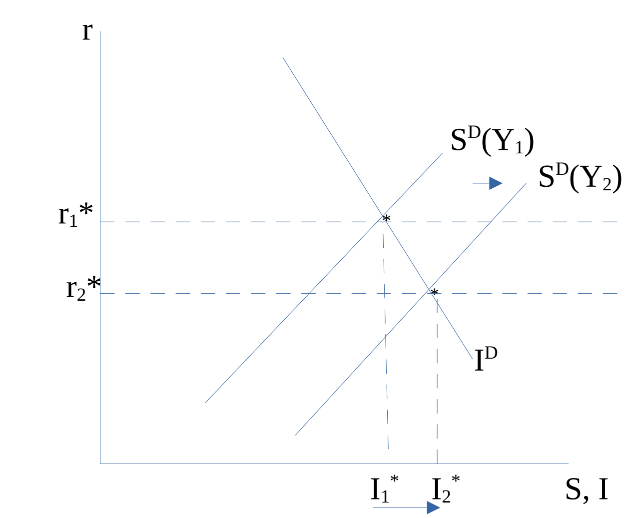
Reasons:
1. Leading indicators are not always consistent predictors of recessions (as noted in part a)
2. Factors that cause recessions may change as economies change –and models need to be updated to incorporate these changes, but without experience (data) on this impacts, these updates can be more difficult.

42. Graph and explain the impact of a positive supply shock (rise in A) on N*, w*, Y*, r*, P & I*, assuming a market clearing long run framework (no sticky wages/prices).

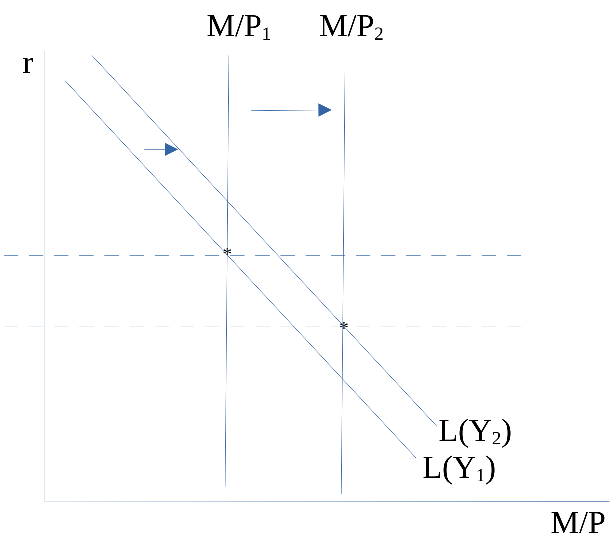
Labour Market



Goods Market



Asset Market



Explanation: Positive supply shock -->A increases --> MPN increases --> production function pivots up due to A and N^D shifts right due to MPN, --> N* increases & w* falls. Y*=F(A,K,N*) increases due to both A rising and N* rising. Since profit maximizing output (Y*) has risen, aggregate supply will be greater than aggregate demand, so prices will fall until demand equals supply.

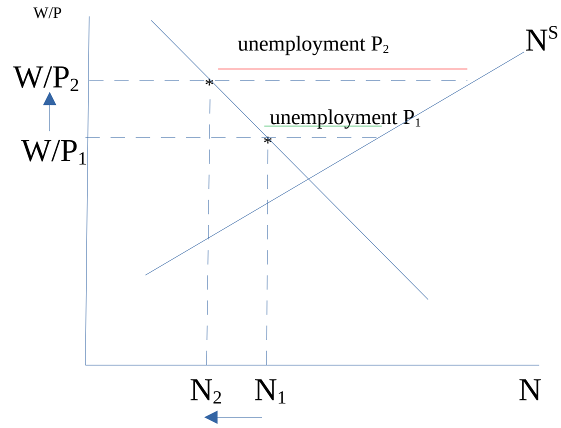
As Y increases, money demand increases (shifts right), S^D increases (shifts right), and r* falls in the goods market. (Note: When r* falls, quantity of consumption demanded increases)

At the same time, the falling P will increase M/P (shifts right). So r* will fall in asset market as well.
You may note that I* rises along side S*, but this is due to a drop in r* (a movement along the I^D curve).

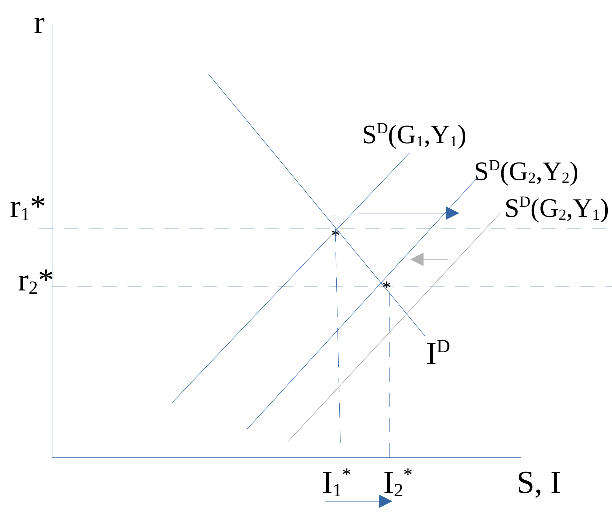
N*: rises
w*: falls
Y*: rises
r*: falls
I*: rises
P: falls

43. Graph and explain the impact of decrease in G on S*, I*, r*, Y, P, N & w in sticky wage framework.

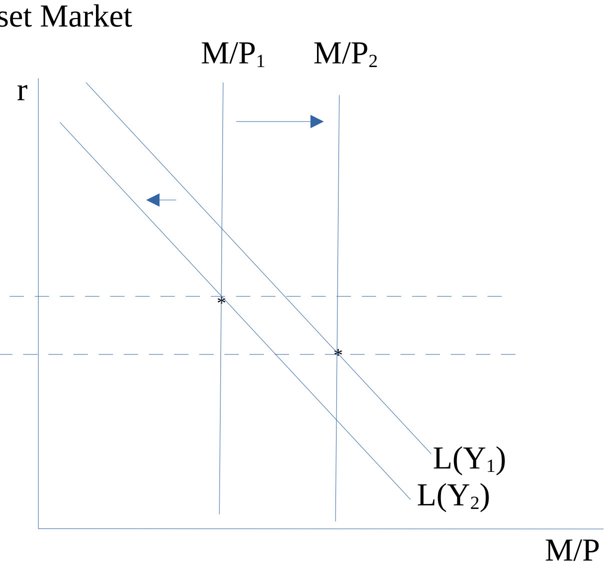
Labour Market



Goods Market



Asset Market



Explanation: G falls --> S = Y-C-G so S rises –> S^D shifts right --> r* falls.

Decreased G - -> $Y=C^D+I^D+G$ falls. Aggregate Demand < Aggregate Supply --> Price falls.

Decreased Y --> L(.) decreases (shift left) & S^D shifts back left a little (not superseding initial impact)
Decreased P --> M/P increases (shift right)
r* in asset market is lower.

Decreased P --> W/P rises --> Employment falls, Unemployment rises.

(Note: When r* falls, quantity of consumption demanded increases as well, but this is moving along the S^D curve as we fall from r_1* to r_2*. Likewise, I* rises along side S*, but this is due to a drop in r* (a movement along the I curve).

N: falls
w: rises
Y: falls
r*: falls
I*: rises
P: falls

48. Answer the following questions on economic growth:

a) List three major input variables associated with growth.

3 input variables associated with growth: 1. Labour (or population) 2. Capital 3. Total factor productivity – A (TFP, technology, energy prices, etc) You could also include Human Capital (Education), and Natural Resources (Land & Non-Renewable Resources)

b) Suppose that the Country of Lomond is identical to the Country of Renfrewshire in all respects except that Lomond has lower GDP due to it's lower levels of Capital stock. So both countries have the same rates of saving and of depreciation, but different levels of capital and output. Assuming that neither Lomond nor Renfrewshire have yet reached their steady state balanced growth path (they remain below it), which country would you expect to grow faster over the next year?

Faster Growth: Lomond
Explanation: At lower levels of K, MPK is higher. So if both countries are saving at the same rate and capital is depreciating at the same rate, then each country is investing in new capital at the same rate. Each increase in capital will, however, result in a greater increase in output (Y) for the country with the lower current level of capital (Lomond) because at lower K their MPK is higher.

c) Discuss two negative aspects of growth, when growth is driven by technological improvement.

Negative aspects of growth: 1. Obsolescence of some existing capital (loss/waste) 2. Adverse effects on some workers/capital owners (those with skills/capital to produce with old technology) – the workers may no longer find their labour or product is competitive with the new labour/production processes, and face unemployment due to this structural shift in the economy. Likewise, firms with obsolete capital will not remain competitive. Given that some workers and capital owners can face sharp declines, there is potential for increasing inequality in the economy when technological improvement leads to large sectoral shifts.

d) Continuing from part c, explain one policy that a government might implement to mitigate the negative aspects of technology driven growth.

Policies to mitigate negative aspects: (you might answer any of these, or potentially others) 1. Training programs (to help workers and businesses develop skills and transition into alternative employment/business opportunities). 2. Subsidizing (via low interest loans) new capital purchases (to help update technology so that firms without sufficient equity or cash flow can still make the transition to new technology and remain in business) 3. Provide generous employment insurance payments to affected workers to mitigate the negative employment effects while these workers transition into different employment (or into retirement).

49. With the increasing capabilities and use of AI (Artificial Intelligence), some individuals have raised concerns over job loss.

a) Why might skilled labour have good cause to worry about job loss with increases in AI?

Cause for Concern:
Skilled workers may have reason to worry about job loss if their occupations are highly likely to be transformed by AI and if there is a low level of complementarity of their skills with AI. In these cases, AI might represent competition for their labour, rather than complementing it, and present a greater risk of job loss (obsolescence). For workers in these occupations, finding and developing complementarity with AI might be a good strategy.

b) Explain one potential benefit to AI for skilled and unskilled workers. Why might skilled labour benefit from AI?

Benefits of AI:
Skilled workers may benefit from AI if there are existing or potential to develop complementarities, because this technology can improve productivity of the workers. Examples include medical professions where this complementary technology improves outcomes.

50. Answer the following questions on measuring poverty:

a) Why is it difficult to define poverty? Explain two reasons.

Difficulties defining poverty:
1. Defining a line below which an individual (or family) is "deprived," is complicated because deprivation is difficult to determine. For example, what amount of money is truly necessary to survive physically (what amount of which food and shelter items –may differ across individuals with different physical characteristics), and participate meaningfully in society (what amount of money is required for meaningful participation – may differ across societies in terms of how participation is conducted along what platforms/venues & may differ across individuals in terms of what is meaningful integration for them).
2. There are multiple measures which offer alternative perspectives on poverty (e.g., income, wealth, consumption), each of which provides a different perspective (consumption may be a useful summary of poverty because it would not treat someone as poor simply because they have low income currently but they have sufficient wealth -or expectations of future income- such that they can dis-save/borrow and smooth their consumption during their periods of low income. However, it is harder to obtain/calculate measures of consumption across the population and consumption is a difficult measure as well because of the level of discretion involved - which cycles us back to the first issue).
Other reasons you might alternatively state could be elaborations or specificities of any of the above, plus concerns about certain measures not adequately capturing re-distribution of income or provision of public goods.

b) Explain why poverty is a subjective and a relative measure.

Poverty is Relative because: As poverty is about deprivation of the resources not only to survive physically, but also to meaningfully participate, there is an element of relativity in that meaningful participation must take into account how a society participates, and therefore the cost of that participation will depend on how society participates – which means it will be relative. For example, if a society has developed and purchased goods in a way that meaningful participation in that society involves access to media, entertainment, information sharing, and other content that requires the use of a television or computer, then someone who does not have sufficient funds for use of a television or computer would be poor. In this way, the poverty line would be relative to what the rest of society is able to purchase with their income/wealth. The relative nature of poverty is also reflected in the understanding that costs and public goods provisions vary across jurisdictions, so, for example, what is poor in Ontario is different from what is poor in Quebec.
Poverty is Subjective because: People have different views on the amount of resources necessary for meaningful participation (or even sufficient nutrition/shelter). Moreover, physically, different people will have different nutritional and shelter needs.

51. In 2018, the Canadian Government initiated a poverty reduction strategy based on the Market Based Measure (MBM) of poverty. Suppose the poverty rate is measured as the percent of families with family income below the poverty line as measured by the MBM.

a) Suppose that the number of families in Canada remains constant between 2025 and 2026, and that the poverty line remains constant between 2025 and 2026. Further suppose that 100,000 families with income below the MBM in 2025 experience a further decline in income in 2026, but no other families in Canada experience a decline in income, would the poverty rate change in Canada? Explain why or why not. (Do in class some examples like this – both decrease and increase in income).

Will the poverty rate change?: No.
Explanation: If these 100,000 families are already below the poverty line (and the poverty line remains constant), then a reduction in their income will not change the fact that their income is still below the poverty line in 2026. Since no other families in Canada are experiencing a reduction in income, no other families will be falling below the poverty line in 2026. Given that poverty rate is the percent of families below the line, the poverty rate= 100 x (# of families with income below line) ÷ (Total # of families) And since the numerator (the # of families with income below the line) is not changing, nor is the denominator, then the poverty rate will be unchanged in 2026.

b) Suppose poverty rates fall by 10% in 2027, and the government claims that this decline indicates the effectiveness of the poverty reduction strategy. Explain what concern(s) you might raise about this claim.

Concerns: A drop in poverty rates doesn't tell us what happened to the families that are no longer in poverty. Yes, the poverty reduction strategy may have helped some families rise above the poverty line, but I would be hesitant to call the strategy effective unless I know how those families are faring. Did their incomes rise just $1 above the poverty line? Are they at risk for falling back below the line? Etc. If the poverty reduction was due to minimal changes in income which did not substantively improve the wellbeing of these families – I would consider whether other strategies might be more impactful (per $ input). Moreover, if these strategies only offered temporary income relief – and long run these families remain at significant risk to fall below the poverty line, then I would not agree with this claim of effectiveness. These concerns are among the many reasons why it is important to consider poverty (or inequality) taking into account more than one measure.

c) In addition to income based measures of poverty and inequality, economists consider additional measures and dimensions. List three alternative measures/dimensions which are important to consider for a fuller picture of poverty and inequality. Describe why each are important.

3 alternative measures/dimensions: Alternative measures/dimensions 1. consumption based 2. wealth 3. education based measures 4. measures of depth 5. measures of duration
Explanation: Income based measures of poverty don't take into account the ability of higher wealth families to smooth their consumption during periods of low income. Wealthier families may dip into savings or borrow to smooth consumption. So alternative measures that consider wealth or consumption can provide a more fulsome assessment of poverty and inequality. Measures that consider education also can be useful because education speaks to long run earnings potential that can, again, be used for consumption smoothing. Finally, when considering poverty or inequality, measures that account for depth are important –if the majority of families in poverty are just below the poverty line, this is a very different situation (and might require different solutions) if these families are substantially below the poverty line. Likewise, if poverty is persistent for these families, this represents a larger problem that might require strategies that address long-term factors influencing income and wealth prospects and/or reliance on a system not well-designed to support these long-term considerations.

52. List two benefits and two drawbacks of using generalized entropy measures (e.g. Theil index) rather than the Gini to measure inequality.

3 alternative measures/dimensions: **Benefits:** 1. rank preserving 2. invariant to population size 3. invariant to scale 4. additively decomposable
Drawbacks: 1. less intuitive than Gini or the decile ratios 2. not directly comparable across populations with different sizes/groups 3. different weight structures can make the measures more sensitive to changes at tails of distributions

VIII. Monetary and Fiscal Policy

53. Answer the following questions on monetary and fiscal policy:

a) Describe two reasons economists might support the rules approach to monetary policy

Drawbacks: 1. Difficulty in predicting future (particularly given lags) mean that a set rule may provide more stability than ad hoc decisions. (Although as predictions get better, some discretionary policy may be effective). 2. A standardized or rules approach can improve credibility of central bank, increasing the effectiveness of the policy (more able to reduce inflation without triggering a recession).

b) Discuss three limitations to conducting monetary policy

Drawbacks: 1. Lags between policy implementation and impact on the economy can be long, so it is difficult to know whether expansionary or contractionary policies will be needed in the future when the policy's effect is felt. 2. Monetary policy has implications on exchange rates, and therefore international trade and investment. Targeting these issues would limit the central bank's ability to target inflation using monetary policy. 3. In the long run, when prices adjust, monetary policy would not have a lasting impact on GDP or growth. (Outside of potential impact of stability on growth, monetary policy may not be as effective for mitigating economic contractions)

c) Describe two drawbacks of fiscal policy:

Drawbacks: (one or more of the following)

1. Distortion (Government spending to implement fiscal policy requires taxation, which has a distortionary effect on the economy -->inefficiencies)
2. Deficit (Government spending which is not paid by taxation results in deficits, which add to debt and increase costs to servicing the debt).
3. Inflexibility (it is difficult/time-consuming to increase G or tax, for most forms of government spending)
4. Lags between implementation of policy and it's effects on the economy can be lengthy, and therefore it is difficult to predict how much G is needed where and when.
5. Increases in G can lead to inflation.

54. Suppose the new government of Spartinth is considering three policies which would increase Government Spending: 1. building new roads to facilitate traffic, 2. a food relief program -allocating funds to people whose income falls below a specific threshold, 3. Provision of a lump sum increase in funding to public universities for the current year.

a) Which policy/policies represent an automatic stabilizer and why.

Automatic Stabilizer(s): #2, the food relief program

Explanation:
The food relief program represents an automatic stabilizer because in economic down turns, fewer people will be employed, and of those employed, many could be earning lower incomes. Therefore, more people will have incomes falling below the threshold and eligible for the funding. The government will, therefore, be spending more money (increased G) during economic downturns. Conversely, when the economy is booming, incomes are higher, so fewer people will be below the threshold and eligible for the funding, so government spending via this program will be lower. Policy 3, is a lump sum one time amount, so it is not an automatic stabilizer. Likewise, policy 1 is a one time increase in G (although the roads may require repairs in the future, these are not automatically increased/decreased with economic contractions and expansions).

b) Do any of these policies have potential negative effects on economic growth or fluctuation? Explain.

Answer:
Policy #2, as an automatic stabilizer could smooth economic fluctuation, and if the effect is a less deep or less lengthy recession (or firms going out of business), could prevent a decline in long-term growth. Policy #1 is unlikely to affect fluctuations as it is a one time event, although it could result in a one time expansion. This investment in infrastructure could, however, increase future growth by facilitating transportation and business development in the country. Likewise, Policy #3 is unlikely to affect fluctuations outside of a one time expansion. However, it could increase future growth if this funding enabled the public universities to provide more/better training of students – improving future labour productivity.

IX. International Economic Concepts, Exchange Rates and Balance of Payments

55. Individuals sometimes raise concerns about the trade deficit being "too large" or "too persistent."

a) What is the trade balance? And how does it differ from the Current Account balance?

Trade Balance: The trade balance is the difference between exports and imports, that is Exports-Imports. So a country's trade balance is their NX. A negative trade balance means the country is importing more than they are exporting.

Difference between Trade Balance and CA balance:

The trade balance is NX, where as the CA includes NX + NFP. So the CA is a broader measure that includes not only the trade balance, but also foreign aid and international investment.

b) If a country has a negative CA balance, is it a net borrower or a net lender?

Answer: Net Borrower.
Explanation: A country with a negative trade balance is importing more than they are exporting, therefore must borrow (sell assets) in order to fund the difference.

c) Explain why a negative CA balance can be a good thing.

Benefits:
A negative trade balance can occur when productivity in a country is high or expected to be high/higher. This expected high productivity means that it is worthwhile to invest. Therefore, net borrowing (negative CA balance) and high levels of investment in this country may be an optimal decision. Borrowing to invest in future production also increases ability to repay the debt in the future.

d) Describe one concern with having a persistent negative CA balance?

Concern: (one or more of these)
1. In general, a negative CA balance may be concerning if it is driven by excessive consumption of Imports, as opposed to being driven by investment in the country's future production. (Imports of "inputs to production" e.g. to support future domestic production should be treated separate from imports of consumption items).
Other concerns you might raise:
2. Persistent negative CA balances may be of concern if the returns to production are not as high as expected and/or insufficient to service the interest+principle on the borrowing.
3. Moreover, Persistent negative CA balances may leave the country more exposed to economic and financial shocks in the future which may limit their ability to service the interest+principle.
4. Persistent negative CA balances may also be of concern if the country will need to raise taxes in the future to pay for it (taxes being inefficient, in addition to the concern of distribution –who & which generation is paying for the current deficits).
High levels of accumulated debt by any one country can impact other country's willingness to take on the risk of lending to that heavily debt burdened country.

56. Many countries with low GDP have borrowed heavily. This borrowing, as discussed in question 57c), can be a good/optimal decision. However as noted in the press, many such borrowing countries have experienced debt crises, where some countries have been unable to make payments, and where the cost of servicing their debt can represent a significant portion of their GDP (and in some cases represents 20-40+% of government budgets) and hinder development.

a) Explain one factor that can contribute to debt crises for small open economies.

Factor: (there is more than one, a couple are provided below). *A rise in real interest rates. Changes in global interest rates are beyond the control of the small countries in debt. A rise in real interest rates (just like a rise in the mortgage rates) can increase the risk of default for these countries as they will struggle to service debt payments under higher interest rates. (Note that many countries are looking at payments that represent a substantial portion of their government budgets, the higher this fraction, the greater the probability that they will not be able to make their payments and maintain basic government services and functions). *Changes in world demand for their products Again, these changes are beyond the control of the small countries in debt. Increasing production revenue and exporting their products is an important means to pay down the debt (and not incur further deficit). If global demand for their exports (and thereby the prices they can command on the global market) are falling, they will have less income with which to pay interest and principle on the debt.

b) Explain how recent increases in inflation, globally, have contributed to the debt crisis for these small open economies.

Explanation: With the recent increases in inflation, many central banks have increased policy rates, leading to a rise in interest rates globally. As discussed above, higher interest rates are a key factor in debt crises because it decreases the borrower's ability to service the debt.

57. For each of the following 'shocks' to an economy, explain the impact on exchange rate for that country:

a) Large reduction in labour supply (relative to other countries) following the COVID pandemic.

Impact on e: rise
Explanation: A decrease in labour supply (shift left of N^S curve) will reduce N* and Y*. The drop in Y* will reduce demand for imports, increasing NX (shift right) (Note saving will also drop with fall in Y causing S-I to shift left) -->r* rises. A drop in Y which increases NX will increase e (because exports>imports --> increased relative demand for this country's currency). A rise in r* will increase desire to invest in this country, raising dollar demand, and increasing the exchange rate e.

b) A reduction in Government Spending following the end of pandemic supports.

Impact on e: ambiguous
Explanation: A drop in G will result in an increase in Saving (rightward shift of S-I) since S=Y-C-G, and a decrease in the r* that clears the goods market. GDP = Y = C+I+G+NX will fall because G has fallen. The fall in r* will reduce desire to invest in this country, thereby reducing dollar demand and reducing e. The fall in Y will result in fewer imports (a small rise in NX) which lowers supply (shift left) relative to demand of this country's currency –raising e. Therefore the impact on e is ambiguous and would depend on whether the r* or the Y effect dominates.

58. Relative to the USD, the Canadian dollar has fluctuated but overall has been declining for about two decades. Some suggest that this decline is a good thing, and that the Bank of Canada should target a lower exchange rate in order to increase our exports and decrease our imports.

a) Explain how the Central Bank could aim to lower the exchange rate for the Canadian dollar. What policy could they use?

Policy: expansionary monetary policy
Explanation: Expansionary monetary policy --> lower r* --> less demand to invest in Canada, less demand for Canadian dollars --> lower e.

b) Do our models suggest that exports increase and imports fall (NX rise) as a result of this policy?

Is NX predicted to rise: not necessarily
Explanation: The drop in the exchange rate could stimulate demand for Canadian exports, increasing NX. (M rises, r* falls, e falls, NX rises – along the curve). However, this effect would be mitigated by the increase in Y arising from the expansionary monetary policy --> increasing imports and lowering NX. So theoretically, the impact on NX is ambiguous. (Note: there are several mitigating effects arising from feedback loops into the system, but these will typically be smaller).

c) What is the potential impact of this expansionary Monetary policy on foreign investment in Canada?

Predicted impact on Foreign investment: ambiguous
Explanation: While the drop in the exchange rate could stimulate foreign investment because it is now cheaper to buy Canadian dollars –there is a reason why it is now cheaper to buy Canadian dollars. The expansionary monetary policy generated a fall in r* which reduced the return on investment, which decreased desire to invest in Canada. Moreover, if the impact on NX were to increase as proposed in the initial question, then overall exports would be rising more than imports –relatively more foreign lending than borrowing – not the other way around.

59. Would the Bank of Canada be able to reduce the exchange rate and reduce inflation simultaneously using an expansionary monetary policy? Explain.

Answer: No
Explanation: An expansionary monetary policy would be required to reduce the exchange rate. But an expansionary monetary policy reduces r* and stimulates aggregate demand. Of course, since productivity is unchanged then optimal production is unchanged. So demand will exceed supply and result in price increases (expansionary monetary policy --> inflation). Therefore the expansionary policy used to reduce the exchange rate will not help to reduce inflation.

60. Suppose that your have data on USD exchange rates, as well as data from the Economist's "Big Mac Index,"‡ across several countries. Suppose that this report shows the price of a Big Mac in South Korea as 5200 Won. While the price of a Big Mac in the US is 5.58.

‡ The Big Max index shows the price of Big Macs in the local currency of several countries. The ratio of 'the price in local country in the local currency' to the 'price in USD in the US' generates the implied purchasing power of the USD in the local country.

a) What is the implied purchasing power of the USD in South Korea?

Answer: 931.9
Explanation: The implied purchasing power of the USD in South Korea is the cost of buying a Big Mac in Wons in South Korea, relative to the cost of buying a Big Mac in USD in the US. This is 5200/5.58 or approximately 931.9.

b) If PPP holds, what would be implied the exchange rate be for South Korea?

Answer: 931.9
Explanation: Purchasing Power Parity (PPP): similar foreign and domestic goods should have the same price in terms of the same currency, so we should see e=1 (real exchange rate is 1:1). If PPP holds, then the nominal exchange rate should equal the price in South Korea (in Wons) divided by the price in the US (in USD). That is 5200/5.58 or approximately 931.9 (Won per USD). To buy 1 Big Mac in the US takes 5.58, which, if the exchange rate were 931.9 would generate about 5200 Won, enabling you to buy 1 Big Mac in Korea. So the exchange rate implied by PPP is 931.9.

c) If the actual exchange rate is 1,274, does the Big Mac Index data suggest that the Won is overvalued or undervalued?

Answer: Undervalued
Explanation: The actual exchange rate suggests you require more Won to buy 1 USD, than is implied by the purchasing power estimated using the Big Mac Index. Specifically, the difference is 343 Won. Which implies the Won is undervalued by close to 27% (100*343/1274).

d) Explain two drawbacks to using the Big Mac Index to estimate whether currencies are over/under valued.

Drawbacks: 1. Big Macs are not the same across countries. Despite general similarities, they may differ in terms of size, ingredients, nutritional content, and what is included in the price (e.g. ketchup, etc). 2. Transportation/input costs of ingredients may vary across countries, as well as market demand, which can result in variation in localized prices.

e) Consider prices of diamonds, bananas, and cars across countries. Which price index measure might be more useful than the Big Mac to estimate over/under valuation of currencies, and why?

Answer: Diamonds have the advantage of being light and cheaper to transport, as well as being close to a final product (less complicated), and having more fixed standards for valuation Bananas are slightly heavier, rot easier, and therefore transportation is a bigger factor. Cars are much heavier, more costly to transport, and involve a more complex and international manufacturing aspect. Therefore a price index comparison for diamonds would make a better index to measure under/over valuation of a currency than Cars or Bananas.

61. Suppose that Mexican pesos depreciate in value relative to Canadian dollars. Specifically, suppose that in 2007 one dollar was worth 10 pesos, whereas in 2027 one CAD could purchase 15 pesos. Lets simplify things and assume that the only product produced in Canada and Mexico are churros.

a) If the price of a churro in Canada is $1 in 2007, and 1.50 in 2027, what is inflation in Canada over this period? Show your calculation.

Answer: 50%
Calculation: 100*(1.5-1)/1

b) If the price of a churro in Mexico was 5 pesos in 2007, and 15 pesos in 2027, what is inflation in Mexico over this period? Show your calculation.

Answer: 200%
Calculation: 100*(15-5)/5

c) Does the depreciation of the Mexican Peso mean that it is cheaper for Canadians to visit (and eat) in Mexico in 2027 compared to 2007? Explain.

Answer: No. It is not cheaper.
Explanation: In 2007, $1 could buy 1 churro in Canada. To buy one churro in Mexico would require 5 pesos, which would cost 50 cents at the 2007 exchange rate. So 1 churro in Mexico cost 50 cents in 2007. So the price of a churro in Mexico is about 1/2 the price of a churro in Canada. In 2027, it costs $1.50 to buy 1 churro in Canada. To buy one churro in Mexico would require 15 pesos, which would cost 1 dollar in 2027. So one churro in Mexico costs $1 in 2027. So the price of a churro in Mexico is now 2/3 of the price of a churro in Canada. Even though it is cheaper to buy churros in Mexico in both 2007 and 2027, it has become relatively more expensive to buy churros in Mexico in 2027 relative to 2007. This is because Mexico experienced greater inflation than Canada, so much so that it surpassed the effect of the devaluation of the Mexican Peso.

62. Suppose the world is comprised of two large, open economy, countries: Florin and Guilder. And suppose that country Florin currently had a negative trade balance (imports>exports).

a) Is Florin a net lender or a net borrower in the world?

Answer: Net borrower.
Explanation: They are importing more than they are exporting, therefore must borrow (sell assets) in order to fund the difference.

b) If future marginal product of capital rises in Guilder, explain what will happen to the trade balance in each country.

Answer: Trade balance will fall in Guilder. Trade balance will rise in Florin.
Explanation: An increase in MPK^f in Guilder will increase the profit maximizing K* and increase investment demand in Guilder. This increased investment demand puts upward pressure on the real interest rate (as I rises, S-I falls, and r* will rise). As r rises, NX fall. So for guilder, the trade balance will shrink, and could even become negative (that is Guilder could become the net borrower in the world –which makes sense if their MPK^f is high –borrow & invest). Because NX in Guilder is falling, NX in Florin is rising, therefore their trade balance is increasing (it could even become positive) (Note that some of the increase in investment in Guilder's capital can actually come directly from abroad -that is from Florin)

Note: Therefore if marginal productivity (of capital) is rising in a set of countries, we would expect that country to borrow more . This increased borrowing would raise r* globally and therefore we would expect all the other countries countries to decrease their own borrowing. One suggestion some economists have made is that if the MPK^f in countries other than the US were to rise, then the US trade deficit might begin to reverse.

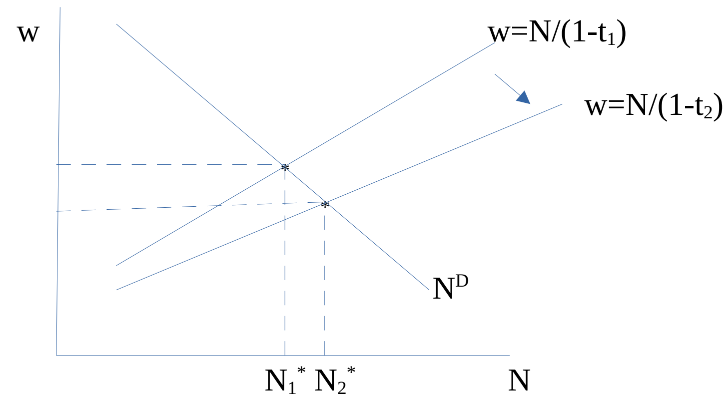
62. Suppose that there are only two countries in the world: Nelarn is a small open economy, and Orr is a large open economy. If Orr reduces taxes on investment income and Nelarn does not, what will be the impact on the net exports in Nelarn? (For this question, abstract from any secondary effects e.g. due to potential changes in exchange rates)

Answer: NX will rise in Nelarn.
Explanation: a reduction in tax on investment income in Orr will increase investment demand (the S-I curve will swivel up), increasing r* and reducing NX. A reduction in NX for Orr means an increase in NX for Nelarn.

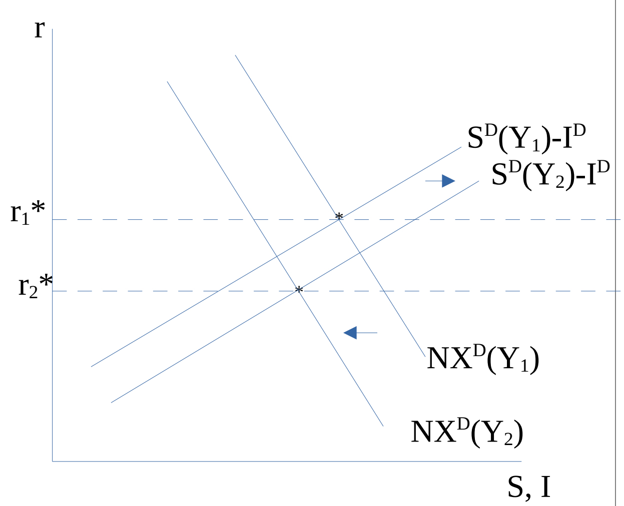
63. The film “Commanding Heights” claims that Ronald Reagan's tax cuts got the U.S. economy out of stagflation (stagnant economy & inflation at same time).

a) Explain how a reduction in income tax, t (where $N^S=(1-t)w$) could reduce inflation and real interest rates, and increase employment and production in a market clearing long run framework. Use the Labour Market, Asset Market and Goods markets diagrams to help explain how the “Commanding Heights” assessment could be correct.

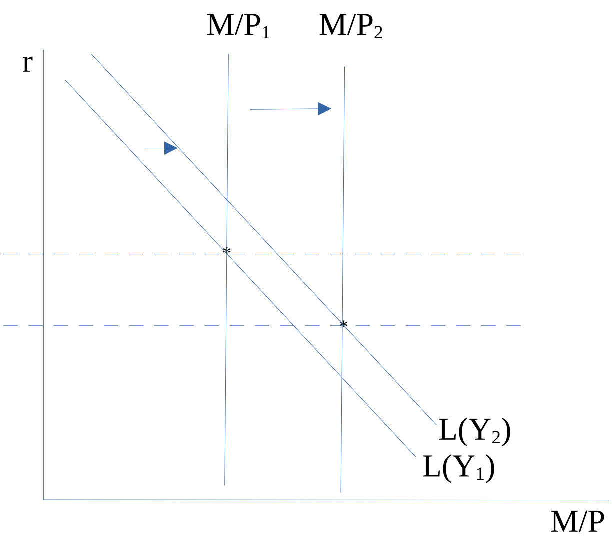
Labour Market



Goods Market



Asset Market



Explanation: A decrease in t, increases the denominator of the labour supply curve, decreasing the ratio, thereby decreasing the slope (see graph above). The downward pivot of the labour supply curve represents an increase in labour supply at any given wage. Equilibrium employment increases to N_2^*. As full employment level of employment rises, so too does full employment level of output ($Y^*=AF(K,N^*)$ increases). The increase in profit maximizing Y* means supply will exceed aggregate demand so prices will fall (until supply = demand)

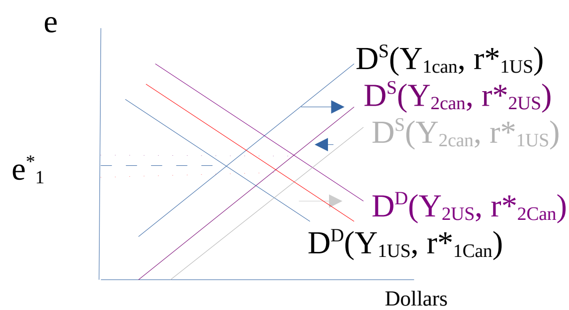
Now, as Y increases, saving demand increases since $S^D= Y – C^D – G$, so $S^D - I^D$ curve shifts right. With higher levels of Y imports also rise, so NX decreases, and the NX curve shifts left. r* falls.

(Note that consumption will increase, but since consumption is a fraction of Y (mpc), it won’t rise by as much as Y. We are asked to assume below, that the Y_{US} effect dominates NX between US & Canada, so these graphs show NX decreasing for the US as they will be increasing for Canada.)

Also note that with higher Y, money demand increases, L(.) shifts right. And as price falls, M/P rises, so r* falls.

b) Assume that U.S. net exports fall as a result of Reagan's policy (Y effect dominates). Explain the impact of Regan's policy on Canada's NX, GDP and e. (Assume US only trades with Canada)

Impact on the following variables (underline one):	**Explanation:**
NX (increase, decrease, same, ambiguous)	US increase in Y effect dominates: Higher Y increases US imports-->lowers US NX --> Canada's NX rises since US buying more Canadian exports.
GDP (increase, decrease, same, ambiguous)	Y=C+I+G+NX, so the increase in NX will increase GDP in Canada
e_{nom} (increase, decrease, same, ambiguous) increases if I said US Y effect dominates.	The increase US GDP and resulting increase in demand for Canadian exports, will increase demand for Canadian dollars. Moreover, the increase in Canadian NX (rightward shift) will increase r* in Canada. With a higher r* more people want to invest in Canadian markets and need to buy Canadian dollars to do so, increasing demand for Canadian dollars this way as well. Also, the lower r* in the US will cause Canadians to want to invest less in the US, reducing supply of Canadian Dollars which decreases D^S, each of these effects would increase e. However, the higher Y in Canada will increase imports demanded, increasing selling of Canadian dollars (to buy USD) --> So a rightward shift of D^S Reducing e. The final effect on e is ambiguous, and depends on whether the increased demand is larger (purple) or smaller (red) relative to the increased supply (purple). Generally the effects arising from the larger economy dominate empirically, but the true answer is ambiguous as it can theoretically go either way depending on whether the demand shift is relatively larger or smaller than the supply shift effect. If the Y_{US} effect dominates the exchange rate outcome, then e_{nom} will rise.



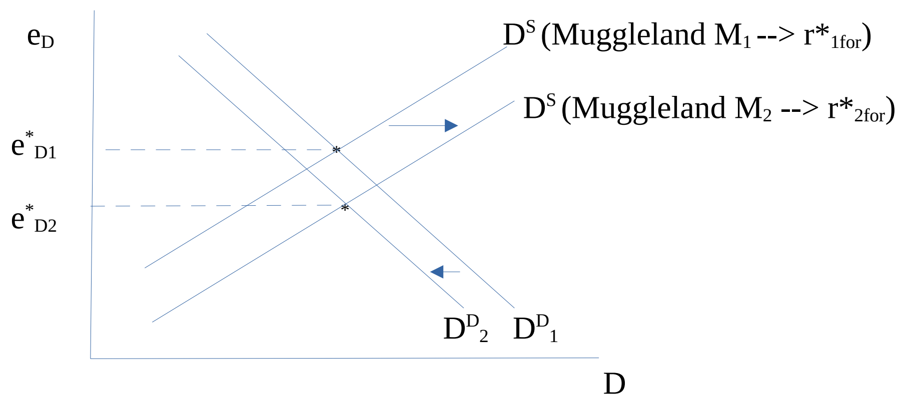
c) Will the trade balance & exchange rate always move in opposite directions? Explain why/why not.

Answer: No
Explanation: The trade balance (NX) and exchange rates (e) do not always necessarily move in opposite directions. Both NX and e are influenced by many factors, some of which may result in ambiguous outcomes. E.g., as shown in part b) Canadian NX rose, but without knowing which effect dominates, the net impact on e_{CAD} is ambiguous. (Likewise for the US – NX fell, but the net impact on e_{USD} is ambiguous – the US increase in Y with subsequent drop in NX would reduce e_{USD}, but the rise in r* in the US would bolster e_{USD}).

64. Suppose the large open economy of Muggleland uses a contractionary monetary policy resulting in lower Y in Muggleland. Assume the Y effect dominates changes in NX in Muggleland and abroad.

a) How will Muggleland's monetary policy impact the economy of their trading partner Diagon Alley (a smaller country)? Specifically, how will Diagon Alley's Y, r, NX and e change? Use the a diagram for Diagon Alley's exchange market to help you explain what happens to e_D (where e_D represents the exchange rate for Diagon Alley's currency relative to the currency of Muggleland). Let D^S and D^D represent demand and supply for Diagon Alley's currency(dollars).

Diagon Alley's Exchange Market



Answer: Diagon Alley will have lower Y, NX, r*, and e*

Explanation: Contractionary monetary policies have recessive effects on the economy. A contractionary monetary policy in Muggleland will reduce Muggleland's Y and increase their real interest r* (leftward shift of M/P, Asset market clears at higher r. Higher r ->lower C, lower Y).

With a lower Y, Muggleland will demand fewer imports (from Diagon Alley), so Diagon Alley's exports fall and NX decreases in Diagon Alley. With NX falling, then Diagon Alley's Y also falls (Y=C+I+G+NX). Note that the decrease in NX for Diagon Alley is represented by a leftward shift of the NX curve – resulting in a lower r* for Diagon Alley.

As Muggleland is demanding fewer imports from Diagon Alley, then demand for Diagon Alley's currency decreases (leftward shift of D^D seen above). Moreover, because r* decreased for Diagon Alley, fewer people will want to invest in Diagon Alley and demand for Diagon Alley currency will fall (shift left) for that reason as well. With lower demand for Diagon Alley dollars - the equilibrium exchange rate will be lower.

But there's more.
Recall that Muggleland's r* had risen due to the contractionary monetary policy. This higher r* will increase desire to invest in Muggleland, so folks in Diagon Alley will want to buy Muggleland currency in order to invest. They will sell Diagon Alley dollars to buy Muggleland currency, so Supply of Diagon Alley dollars increases (shifts right) --> lower equilibrium exchange rate.

Now Diagon Alley's lower Y and lower exchange rate will have a secondary effects -- reducing imports which would boost NX somewhat. But recall that we are asked to assume that Muggleland's Y effect dominates, so NX remains reduced in Diagon Alley.

b) Now, instead of allowing e to fluctuate, assume that these countries were in a fixed exchange rate regime and Diagon Alley pegged their exchange rate to Muggleland. (Assume they want to keep it pegged at an official rate equal to e^*_{D1}). Would the impact of Muggleland's policy on Diagon Alley's GDP (Y) be larger or smaller? Explain.

Answer: Larger.
Explanation: Recall that due to the drop in Muggleland's Y, they reduced demand for Diagon Alley's exports, and NX fell for Diagon Alley. But there was a mitigating effect – with a lower exchange rate Diagon Alley's NX increased slightly (not enough to offset the initial decline, but somewhat). If Diagon Alley pegged their exchange rate to Muggleland's currency, then e_D would not have fallen, and the mitigating effect (that slight increase in NX) would not occur. Therefore, by pegging their exchange rate, Diagon Alley is eliminating this mitigating effect, and the decline in Diagon Alley's NX would be larger, and therefore their drop in Y would be larger.

c) Would your answer from part b depend on how Diagon Alley fixed it's exchange rate? Explain.

Answer: Yes
Explanation: If Diagon Alley pegs their exchange rate by selling official reserve assets (Muggleland currency) and buying Diagon Alley currency to artificially boost e_D, then the only impact of this pegging is the one discussed in part b (loss of the mitigating effect of the exchange rates on NX and therefore Y). {We are not discussing impact on official reserve assets here}. BUT, if Diagon Alley were to peg their exchange rate using Monetary Policy – there are further repercussions on GDP. Given that market clearing e_D (the fundamental value) had fallen below the official rate that they want to peg (e^*_{D1}) then Diagon Alley would have to use a contractionary monetary policy to peg it at that official/original rate. (See lecture slide X). A contractionary monetary policy has a recessive effect on the economy (as discussed above) and would further decrease Y in Diagon Alley. Therefore, it matters which method Diagon Alley uses to fix it's exchange rate. Using contractionary monetary policy has an additional recessive effect on the economy.